

The semantics of anaphora (handbook article) *

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May 28, 2025

Abstract. This chapter explores the semantics of anaphora by highlighting analytical challenges posed by anaphora with unspecific antecedents and reviewing solutions proposed in semantic theory. The main focus is how pronouns establish reference in hypothetical and counterfactual discourse, contrasting descriptive and dynamic approaches in formal semantics. It provides a detailed discussion of dynamic approaches to the challenges in tracking referents across hypothetical and counterfactual discourse contexts. The discussion also connects to philosophical notions of reference and includes a review of psycholinguistic experiments investigating the cognitive mechanisms involved in interpreting anaphora.

1 Introduction

Anaphora is a dependency between expressions in discourse. A typical example is (1).

- (1) *Mary owns a car_x. It_x 's red.*

The pronoun *it* is an anaphor that depends on its antecedent *a car* for its interpretation. Here, this dependency is one of *coreference*: Both expressions denote¹ the same object (indicated by coindexation). However, defining anaphora based on coreference is problematic, as many cases typically classified as anaphora are non-referential. This chapter addresses issues in the semantics of anaphora by considering analytical challenges posed by anaphora with unspecific antecedents, when there is no specific referent, and reviewing solutions to these challenges proposed in semantic theory. The main focus will be on anaphora with intensionally unspecific antecedents in hypothetical or counterfactual discourse. The chapter is intended to be useful to advanced students of linguistic semantics and researchers interested in reading about the formal semantics of anaphora, or its intersection to theories of language processing.

We begin by laying out the central puzzle that anaphora with unspecific antecedents challenge any characterization of anaphora based on co-reference: Traditionally, an expression is considered referential if it denotes a particular object in the world (Frege 1892), excluding cases like (2a) bound anaphora and (2b) donkey anaphora (Evans 1977), which lack a specific referent.

- (2) a. Bound anaphora: *Every actress_x believes she_x deserves more roles.*
 b. Donkey anaphora: *Every farmer who owns a donkey_y feeds it_y.*

*My thanks are due to Patrick Elliott, Morwenna Hoeks, Adrian Brasoveanu, Hans Kamp, and Antje Rossdeutscher for discussion and feedback.

¹We understand *denotation* model-theoretically, as a semantic value in a Fregean/Tarskian sense.

The meaning of *she* in (2a) can be described using a variable bound by the quantifier *every actress*, yielding the following paraphrase: for every individual x in the domain, if x is an actress, then x believes x deserves more roles, captured in predicate logic as:

$$(3) \quad \forall x(\text{actress}(x) \rightarrow \text{believes}(x, \text{deserves-more-roles}(x)))$$

Because the denotation of the pronoun depends on a quantifier that syntactically commands it, it is analyzed as a variable bound by the quantifier (Montague 1973). The pronoun in (2b), similarly, depends on the quantifier *a donkey*, which, in turn, depends on *every farmer*. This can be paraphrased as: for every x , if x is a farmer and owns a donkey y , x feeds y .

Because there is no specific referent in these cases, (co-)reference does not capture all cases of anaphora. King and Lewis 2021 alternatively consider characterizing anaphora as cases where “the interpretation [...] of one expression depends on the interpretation of [...] another,” but also note that this includes non-anaphoric phenomena like lexical disambiguation. Non-referential anaphora like (2) thus complicates a pretheoretic definition and raises analytical challenges.²

Section 2 begins the discussion of these challenges by discussing referential cases (1), and donkey anaphora (2b), which inspired two competing approaches: the descriptive approach (Evans 1977; Neale 1988), and the dynamic approach (Kamp 1981b; Heim 1982; Groenendijk and Stokhof 1991). Dynamic accounts assume that pronouns are variables that establish a relationship to the antecedent by recovering discourse referents that are introduced and manipulated by the information in a discourse. Descriptive accounts assume that anaphoric pronouns establish a relationship by recovering a definite description, comes with a presupposition that a unique referent exists. And both approaches come with distinct responses to the reference problem posed by unspecific anaphora.

Section 3 shifts to the primary focus of the chapter, namely non-referential anaphora with antecedents in the scope of non-veridical operators that are interpreted in epistemic possibilities that are hypothetical or counterfactual, and thus possibly distinct from the actual world. These include cases such as:

- (4) a. Modal subordination (Roberts, 1987, 1989)
A wolf_x might walk in. It_x would eat you first.
- b. ‘Bathroom-disjunctions’ (Evans 1977; Roberts 1989)
Either there isn’t a bathroom_y in this house or it_y’s in a funny place.
- (5) Modal subordination with negated antecedents (Frank 1996)
Fred didn’t buy a microwave oven_z. He wouldn’t know what to do with it_z.

The pronouns in (4) and (5) are not interpreted relative to quantification over individuals like the bound and donkey anaphora in (2). Instead, they are non-referential because they don’t (necessarily) refer to factual objects. While (4a,b) merely invoke the possibility that the mentioned objects exist, (5) explicitly denies the existence of a referent. Here, we distinguish between *quantificationally unspecific* anaphora, which are associated with real-world objects relative to quantificational values, and *intensionally unspecific* anaphora, which do not point to actual objects. The section gives an overview of the empirical patterns: when intensionally unspecific anaphora is, and isn’t, possible (drawing on Karttunen 1976; Roberts 1987; Kibble 1994; Krahmer and Muskens 1995, and Hofmann 2022, 2025).

²This discussion is limited to singular pronominal anaphora like English *it* or *she*.

Section 4 then evaluates the formal ingredients needed for dealing with various types of anaphora in hypothetical and counterfactual discourse, and how these cases of anaphora bear on the comparison between descriptive and dynamic approaches: while descriptive approaches can extend to intensionally unspecific cases more straightforwardly, a dynamic approaches can be motivated due to their greater generality. This section goes on to carefully consider how a dynamic theory of anaphora must be enriched for these cases, providing a detailed comparison of existing analysis, by using four empirical test cases (double negation, bathroom-disjunctions, modal subordination, counterfactual continuations) as tests for any theory of intensionally unspecific anaphora.

In the spirit of bridging semantic theory, philosophy, and cognition, this chapter also connects to the question of reference in the philosophy of language throughout the discussion; and Section 5 surveys reviews experimental studies which ask whether the humans treat anaphoric dependencies as retrieval of semantic values (supporting dynamic views) or of linguistic forms (supporting descriptive views), and ones that investigate the cognitive mechanisms involved in processing intensionally unspecific anaphora.

Section 6 concludes, suggesting that although descriptive analyses handle many cases of intensionally unspecific anaphora with less machinery, an enriched dynamic semantics offers the most unified treatment once temporal, propositional, and counterfactual anaphora, and real-time processing data are all taken into account.

2 Semantic approaches to anaphora

2.1 Referential anaphora

Our main goal will be providing semantic representations to explain how anaphoric pronouns derive their meaning from their antecedents. In basic cases like (1), repeated here, we want to capture a coreference relation, where anaphor and antecedent denote the same real-world object.

(1) *Mary owns a car_x. It_x’s red.*

Our starting point is the standard Montegovian view, where pronouns are variables that receive their semantic value from a variable assignment function g :

- (6) a. It_x ’s red
b. $\llbracket red(x) \rrbracket^{M,g} = g(x) \in \llbracket red \rrbracket^M$

This allows for analyzing bound anaphora (2a) as quantifier-bound variables in predicate logic (definitions in Appendix A), since quantifiers are interpreted by modifying assignment functions. To extend the analysis to referential anaphora, we can assume that indefinites contribute existential quantifiers (Russell 1919) and take scope over multiple sentences in discourse (Geach 1962):

(7) $\exists x(car(x) \wedge own(m, x) \wedge red(x))$

(7) is an existential statement that is true iff Mary owns at least one red car. Some authors (Evans 1977 *et seq.*) argue that this fails to capture that (1) conveys reference to a specific or uniquely salient car. Others (following Kripke 1977) suggest that specific reference is a pragmatic phenomenon, and existential statements can highlight a specific *speaker’s referent* (Eckhardt 2012

provides an illuminating example of a parent smelling their son, saying: “*Someone needs a bath here*”). We sidestep this issue here³ and note that (7) successfully derives that the pronoun points to the same object as its antecedent, i.e. some x that is Mary’s car.

2.2 The donkey scope paradox

Extending this approach to donkey anaphora (2b) presents a famous scope paradox (Evans 1977; Kamp 1981b; Heim 1982), illustrated in (9), which gives two possible representations (the scope of the indefinite is underlined).

(2b) *Every farmer who owns a donkey_x feeds it_x.*

- (8) a. $\exists x(\text{donkey}(x) \wedge \forall y((\text{farmer}(y) \wedge \text{own}(y, x)) \rightarrow \text{feed}(y, x)))$
 b. $\forall y((\text{farmer}(y) \wedge \exists x(\text{donkey}(x) \wedge \text{own}(y, x))) \rightarrow \text{feed}(y, x))$

Analyzing the antecedent as a wide-scope indefinite (8a), yields the implausible interpretation that there exists (at least) one donkey and all relevant farmers own and feed the same donkey. If we attempt to achieve the unspecific interpretation by letting the indefinite scope below the quantifier (8b), it is no longer in a position to bind the pronoun. That is because the existential quantifier contributed by the antecedent would need to take scope over the pronoun in order to analyze their relationship in terms of variable binding. However, in this case the unspecific interpretation of the indefinite is only possible when the scope of the indefinite excludes the pronoun. Donkey anaphora thus raise the question of how they receive a semantic value.

Moving the indefinite below the quantifier to get the intended nonspecific reading (8b) breaks pronoun binding, because the antecedent no longer c-commands the pronoun. Thus the antecedent seems to require an unspecific interpretation in a position that neither syntactically commands nor semantically scopes over the pronoun, even though such scope would be needed for ordinary binding. Donkey pronouns therefore pose the problem of how the pronoun obtains its semantic value.

This key issue has inspired two prominent approaches: The *descriptive approach* (Evans 1977; Neale 1988) suggests that pronouns do not directly receive their semantic value from their antecedent but instead they recover the predicative content of their antecedents, and reintroduce it as a definite description. The *dynamic approach* (Karttunen 1976; Kamp 1981b; Heim 1982; Groenendijk and Stokhof 1991) posits that anaphora pick up discourse referents introduced in the previous discourse. Below, I briefly introduce both approaches.

2.3 Descriptive approach

The descriptive approach (Evans 1977; Neale 1988; Elbourne 2005) assumes a standard treatment of indefinites as existential quantifiers, but departs from the Montegovian analysis of pronouns. Instead of treating anaphoric pronouns as variables, this approach assumes that they recover the (complex) predicate quantified over in the antecedent (9) and introduce a new, definite quantifier. Accordingly, (10) states that the unique car that Mary owns is red.

³For discussion, see Heim 1982, Ch. 1.

- (9) a. *Mary owns a car.*
 b. $\exists x(\text{car}(x) \wedge \text{own}(m, x))$
- (10) a. *It is red.*
 b. $\text{red}(\iota P)$
 c. $P = \lambda x.(\text{car}(x) \wedge \text{own}(m, x))$
 d. $\iota P = x$, iff $P(x)$ and $|P| = 1$

P represents the predicate recovered from the antecedent. The definite operator ι (10d) presupposes that there is a unique entity satisfying P , and returns that entity (following Russell 1919; Strawson 1950).⁴ While bound anaphora can still be analyzed as bound variables, the descriptive approach represents donkey anaphora as follows:

- (11) a. *Every farmer who owns a donkey feeds it.*
 b. $\forall y((\text{farmer}(y) \wedge \exists x(\text{donkey}(x) \wedge \text{own}(y, x))) \rightarrow \text{feed}(y, \iota P))$
 c. $P = \lambda x.(\text{donkey}(x) \wedge \text{own}(y, x))$

The recovered predicate (11c) contains a free variable y . The pronoun introduces ι as a definite determiner binding x , while y is bound in the scope of the universal quantifier over farmers. The resulting paraphrase is: *For every farmer, if they own a donkey, they feed the unique donkey they own.* Thus, the basic descriptive analysis outlined here successfully derives that donkey anaphora is possible, by allowing pronouns to reuse descriptive content in definite expressions.

2.4 Dynamic semantics

Dynamic approaches (Kamp 1981b; Heim 1982, 1983a; Groenendijk and Stokhof 1991; Kamp and Reyle 1993) treat anaphoric pronouns as variables but depart from Montague’s analysis of quantification. Here, existential quantifiers introduce *discourse referents* (Karttunen 1976)—variable mappings that can be stored and accessed in a representation of the current state of the discourse.

2.4.1 The basic system

Here, we model a discourse context C as a set of variable assignments, and utterance meanings as *updates* to this context. The simple utterance (12a) is represented as (12b), using a version of Dynamic Predicate Logic (DPL, Groenendijk and Stokhof 1991, definitions in Appendix B).

- (12) a. *Mary owns a car.*
 b. $[x]; [\text{car}(x)]; [\text{own}(m, x)]$

The contribution of this utterance is modeled as a series of updates to the discourse state: $[x]$ introduces a new discourse referent (dref) for x . The updates $[\text{car}(x)]$ and $[\text{own}(m, x)]$ impose conditions that x must point to a car owned by Mary. Let’s see how this works in detail.

Variable update. The first update $[x]$ introduces a dref through *variable update* (13).

- (13) DPL variable update: $\llbracket [x] \rrbracket^{M, g} = \{h \mid \exists d \in D: h = g[x \mapsto d]\}$

⁴Descriptive analyses vary regarding proposals of how the description is recovered (e.g., syntactic anaphora like ellipsis, or semantic anaphora over properties) and which quantifier models the contribution of pronouns.

(13) defines a set of output assignments given a model M and input assignment g . It updates the context by randomly and non-deterministically introducing new mappings for x . For illustration, consider a model M_1 , where: $D = \{c_1, c_2, c_3, \text{mary}\}$ is the domain of individuals, mary is the person named *Mary*, c_1 , c_2 and c_3 are cars, and Mary owns c_1 and c_2 . For an arbitrary input assignment g , (13) outputs a set containing all assignments mapping x to an object in D (Figure 1), reflecting the random and non-deterministic nature of variable update.

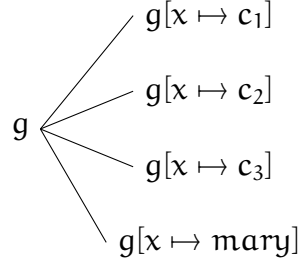


Figure 1: Random variable update

Predication. The updates $[\text{car}(x)]$ and $[\text{own}(m, x)]$ are atomic formulas involving predication (14), and eliminate assignments where x doesn't point to a car owned by Mary, shown in Figure 2.

(14) DPL predication: $\llbracket [\text{car}(x)] \rrbracket^{M, g} = \{h \mid h = g, h(x) \in I_M(\text{car})\}$

Updates not introducing new drefs are tests that eliminate assignments not satisfying the stated conditions. For a single input assignment g , the output of (14) is the singleton set containing the unchanged input ($h = g$) if the condition is met ($h(x) \in I_{M_1}(\text{car})$), or the empty set otherwise.

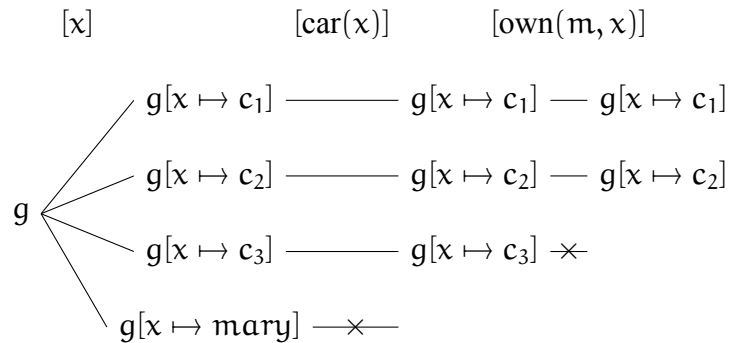


Figure 2: Predication eliminating possibilities

Discourse update and truth. To derive the context change illustrated in Figure 2, DPL updates are applied distributively to each assignment g in the input discourse context C_{in} , and the output context C_{out} is the union of all sets defined by the update:

(15) DPL update: $C_{\text{out}} = C_{\text{in}} + S := \bigcup_{g \in C_{\text{in}}} \llbracket S \rrbracket^{M, g}$

The three updates are conjoined using dynamic conjunction (;). This operation, defined as function composition (16), derives the combined sequence as illustrated in Figure 3.

$$(16) \quad \llbracket [x]; [\text{car}(x)]; [\text{own}(m, x)] \rrbracket^{M, g} = \{g[x \mapsto d] \mid d \in I_M(\text{car}) \text{ and } \langle I_M(m), d \rangle \in I_M(\text{own})\}$$

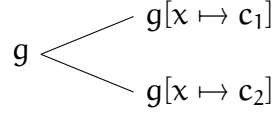


Figure 3: Update with *Mary owns a car*.

In dynamic semantics, the meaning of an utterance is characterized by its *context-change-potential* (Heim 1982), i.e., how it can update available information. A truth-conditional characterization of utterance meaning is given by quantifying over the assignments in the output context. (17) states that a sentence S is true in a context if it specifies an update with at least one output assignment:

$$(17) \quad S \text{ is true relative to model } M \text{ and context } C, \text{ iff } \bigcup_{g \in C} \llbracket S \rrbracket^{M, g} \neq \emptyset$$

An update with (16) yields a non-empty output if Mary owns at least one car in the model, implementing the truth-conditions of existential quantification.

Anaphora. The innovation of dynamic semantics is that the output state now stores a dref x for a car that Mary owns, allowing subsequent pronouns to be interpreted with the same mapping:

$$(18) \quad \begin{array}{ll} \text{a.} & It_x \text{ is red} \\ \text{b.} & \llbracket \text{red}(x) \rrbracket^{M, g} = \{h \mid h = g, h(x) \in I(\text{red})\} \end{array}$$

For M_1 , assume that only c_1 is red. The update with (18) following (16) is illustrated in Figure 4.

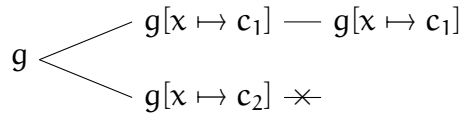


Figure 4: Update with *Mary owns a car. It is red*.

2.4.2 Bound anaphora

Three features of variable mappings in dynamic semantics are crucial for interpreting bound anaphora: (i) they are introduced non-deterministically, (ii) they can be quantified over, and (iii) they may depend on being in the scope of certain operators, such as quantifiers. As in the standard analysis, bound anaphora is analyzed using universal quantification and implication:

$$(19) \quad \begin{array}{ll} \text{a.} & \text{DPL universal quantification:} \\ & \llbracket [\forall x. \phi] \rrbracket^{M, g} = \{h \mid h = g, \forall d \in D: \llbracket \phi \rrbracket^{M, g[x \mapsto d]} \neq \emptyset\} \end{array}$$

b. DPL implication:

$$\llbracket [\phi \rightarrow \psi] \rrbracket^{M,g} = \{h \mid h = g, \forall i \in \llbracket \phi \rrbracket^{M,g}: \llbracket \psi \rrbracket^{M,i} \neq \emptyset\}$$

Universal quantifiers (19a) introduce a “temporary dref” x for interpreting material in their scope. While the condition $h = g$ guarantees that the update is a test, material in its scope is interpreted using an updated assignment $g[x \mapsto d]$, for each element d in the domain. Pronouns in the scope can be interpreted using this variable mapping. Material implication (19b) is also interpreted as a test, requiring that every assignment that results from updating with the conditional antecedent also makes the consequent true. Accordingly, bound anaphora is represented as in (20).

(20) *Every actress believes that she deserves more roles*

$$\begin{aligned} & \llbracket [\forall x(\text{actress}(x) \rightarrow \text{believes}(x, \text{deserve-more-roles}(x)))] \rrbracket^{M,g} = \\ & \{h \mid h = g, \forall d \in D: \forall k \in \llbracket [\text{actress}(x)] \rrbracket^{M,g[x \mapsto d]} : \\ & \llbracket [\text{believes}(x, \text{deserve-more-roles}(x))] \rrbracket^{M,k} \neq \emptyset\} \end{aligned}$$

Because variable update is non-deterministic, x can map to any actress in the model domain. Although the overall interpretation of the quantifier is non-referential (it doesn’t map to a single, specific object), each quantificational instance is referential in a localized sense (for each possible mapping, x maps to exactly one object). The meaning of quantifiers is formalized by quantifying over possible mappings. For (20), we derive the truth conditions that each mapping satisfying the restrictor (x is an actress) also satisfies the nuclear scope (x believes that x deserves more roles).

2.4.3 Donkey anaphora

Donkey anaphora (21) works pretty analogously. Although the antecedent *a donkey* does not syntactically command the anaphor *it*, both lie in the scope of *every farmer* and vary depending on the universal quantifier. The indefinite antecedent introduces a temporary dref in its scope with a specific mapping for each quantificational instance, providing the denotation for the pronoun.

(21) *Every farmer who owns a donkey feeds it.*

$$\begin{aligned} & \llbracket [\forall x(([\text{farmer}(x)]; [y]; [\text{donkey}(y)]; [\text{own}(x, y)]) \rightarrow [\text{feed}(x, y)])] \rrbracket^{M,g} = \\ & \{h \mid h = g, \forall i \in \{g[x \mapsto d_1, y \mapsto d_2] \mid \\ & d_1 \in I_M(\text{farmer}), d_2 \in I_M(\text{donkey}), \langle d_1, d_2 \rangle \in I_M(\text{own})\}: \llbracket [\text{feed}(x, y)] \rrbracket^{M,i} \neq \emptyset\} \end{aligned}$$

The output of (21) retains unchanged input elements ($h = g$), if any possible assignment i that is an update of g with x, y , s.t. x is a farmer and y is a donkey x owns is one that also maps x, y in such a way that x feeds y . Accordingly, an update with (21) is true in a model where every farmer who owns a donkey feeds every donkey they own.

2.5 Reference, accessibility, and taking stock

Basic cases of anaphora support a coreference-based characterization, but non-referential anaphora challenge this view. The descriptive and dynamic approaches offer different solutions to the challenge. In the descriptive approach, coreference may be a by-product of the dependency, which results from retrieving a predicate and introducing a definite quantifier. When this quantifier scopes

high, uniquely identifying a referent, we get referential anaphora. However, when it scopes below another quantifier the pronoun is interpreted unspecifically.

In dynamic semantics, all anaphors involve coreference in some sense, as they inherit their semantic values from the antecedents. The dynamic approach analyzes quantificationally unspecific anaphora by generalizing the concept of (discourse) reference to include a relative sense of reference, picking out a specific object relative to each quantificational instance.

The referential status of drefs in this relative sense depends on the scope of a commanding quantifier. Dynamic semantics predicts that anaphora is acceptable in the scope of the same quantifier (22a), but not outside of it (22b).

- (22) a. *Every farmer who owns a donkey_x feeds it_x.*
 b. *#It_x likes hay.*

This pattern is explained by characterizing the universal quantifier *every* as *internally dynamic* and *externally static* (in the sense of Groenendijk and Stokhof 1991). It is internally dynamic, as drefs introduced in its scope are accessible within that scope, and externally static, because the quantifier is a test, and the same drefs aren't accessible outside of it. Descriptive approaches explain accessibility through presupposition satisfaction: singular anaphora is possible when the definite presupposition of existence and uniqueness is met. For (22b), there is no way that a pronoun outside the scope of the universal quantifier could recover a description that satisfies this presupposition, allowing the descriptive approach to rule out these cases as well.

This section illustrated how descriptive and dynamic approaches address some basic questions about quantificationally unspecific anaphora. There has been extensive discussion in the literature of many related questions, informing evaluations, comparisons, and developments of these approaches. Sidestepping many relevant issues, the remainder of this chapter focuses on analytical challenges posed by intensionally unspecific anaphora.

3 Intensionally unspecific anaphora

This section examines when intensionally unspecific indefinites can be anaphoric antecedents and how veridicality affects anaphoric accessibility. Pronoun use presupposes an existing referent, therefore anaphora is typically blocked with modalized or negated antecedents that do not entail the existence of a referent. However, exceptions occur when the anaphor itself is in a hypothetical or counterfactual context, allowing for local accommodation of the existential presupposition.

3.1 When anaphora is not possible

Karttunen 1976 observed that intensionally unspecific indefinites under negation (23a), epistemic possibility modals (23b), or disjunction (23c), typically cannot serve as anaphoric antecedents:

- (23) a. *Mary doesn't own a car_x.*
 b. *Mary might own a car_x.*
 c. *Either Mary has a car_x, or she walks to work.*
- (24) *#It_x is red.*

Descriptive and dynamic theories can both derive this observation. Under the descriptive approach, anaphora is unavailable because the sentences in (23) do not entail that Mary has a car, and the presupposition of the pronoun isn't satisfied. For (23a), the representations are:

- (25) a. *Mary doesn't own a car* $\rightsquigarrow \neg \exists x(\text{car}(x) \wedge \text{own}(m, x))$
 b. *It is red.* $\rightsquigarrow \text{red}(\iota P)$
 c. $P = \lambda x.(\text{car}(x) \wedge \text{own}(m, x))$

(25b) presupposes that there exists a unique car that Mary owns, but (25a) denies that, creating a contradiction, and thus unacceptability. The dynamic explanation assumes updates with non-veridical operators are externally static tests, that retain only assignments already present in the input ($g = h$). For example, negation and disjunction are given the following semantics:

- (26) a. DPL negation
 $\llbracket \neg \phi \rrbracket^{M,g} = \{h \mid h = g, \llbracket \phi \rrbracket^{M,g} = \emptyset\}$
 b. DPL disjunction
 $\llbracket \phi \vee \psi \rrbracket^{M,g} = \{h \mid h = g, \llbracket \phi \rrbracket^{M,g} \cup \llbracket \psi \rrbracket^{M,g} \neq \emptyset\}$

Content in their scope affects only the informational content of an utterance without introducing new drefs. Dynamic theories thus make a strong prediction that indefinites embedded under non-veridical operators could never provide antecedents for anaphora outside that scope.

3.2 When anaphora is possible

However, the literature also documents various cases of anaphora with antecedents under non-veridical operators. These include modal subordination and bathroom-disjunctions, like (4), repeated here, where the anaphor itself is also embedded under a non-veridical operator.

- (4) a. *A wolf_x might walk in. It_x would eat you first.*
 b. *Either there isn't a bathroom_y in this house or it_y's in a funny place.*

Descriptive analyses predict that these are possible because the pronominal presupposition can be accommodated in the hypothetical local context introduced by the non-veridical operator (see Van Rooij 2006; Keshet and Abney 2024, and discussion in Section 4). Dynamic accounts, on the other hand, are challenged by this data, as they assume that non-veridical operators prevent the indefinites in (4) from introducing drefs. Another counterexample comes from accessible antecedents under double negation (Krahmer and Muskens 1995):

- (27) Krahmer and Muskens (1995): (5)
It's not true that John didn't bring an umbrella_z. It_z was purple and stood in the hallway.

(27) is also ruled out in dynamic semantics by assuming that non-veridical operators are externally static. However, the descriptive approach correctly predicts its availability: (27) entails the existence of an umbrella that John brought, satisfying the pronoun's existential presupposition.

To illustrate how systematically the veridicality of embedding operators affects anaphoric accessibility, this section explores the anaphoric possibilities in veridical, non-veridical, and anti-veridical embedding contexts. For unary operators, veridicality is formally defined as (28).

- (28) Veridicality: [Giannakidou (1998): 20]
 Let Op be a unary propositional operator. Then:
- a. Op is *veridical* iff $\text{Op}(p) \rightarrow p$ is logically valid. Otherwise Op is non-veridical.
 - b. Op is *anti-veridical* iff $\text{Op}(p) \rightarrow \neg p$ is logically valid.

Embeddings under *I know*, *Sue believes*, and *It's not the case* can be characterized as follows:

- (29)
- a. Veridical embedding:
[I know [that Mary has a car]]. ($\rightarrow$ *Mary has a car*)
 - b. Non-veridical embedding:
[Sue believes [that Mary has a car]]. ($\nrightarrow$ *Mary has a car*)
 - c. Anti-veridical embedding:
[It's not the case [that Mary has a car]]. ($\rightarrow \neg$ *Mary has a car*)

Similarly, we distinguish between *factual*, *hypothetical*, and *counterfactual* propositional content, indicating whether an utterance entails an embedded proposition, as in (29a), does not entail it (29b), or contradicts it (29c). We also apply this terminology to DPs like *a car* in (29). and drefs, distinguishing whether the existence of a referent is entailed, not entailed, or contradicted.

3.2.1 Factual antecedents

Factual indefinites which entail the existence of a referent support anaphora even when non-veridical operators are present. Examples include Krahmer and Muskens's (1995) double negation case (27) and indefinites under negated factive predicates (30a) or in negated prejacent of negative implicatives (30b) (Karttunen 1976).

(27) *It's not true that John didn't bring an umbrella_z. It_z was purple and stood in the hallway.*

(30) Karttunen (1976): (16, 14b)

- a. *Bill didn't realize that he had a dime_x. It_x was in his pocket.*
- b. *John forgot not to bring an umbrella_y, although we had no room for it_y.*

Karttunen suggests that anaphora is possible when the speaker commits to the truth of the clause containing the antecedent. This generalization explains the above cases: (27) and (30b) entail that John brought an umbrella, and (30b) presupposes that Bill has a dime. It also aligns with the common intuition that pronouns presuppose that their referent exists, allowing for a restatement:

- (31) **Generalization about accessibility** (preliminary, adapted from Karttunen 1976)
 An indefinite can be an anaphoric antecedent only if its referent is entailed to exist.

This restatement also applies to specific indefinites. For example, under a specific interpretation, the indefinite in (32a) takes scope over negation.

- (32) Context: Mary is enrolled in a class with a list of required readings.
- a. *Mary didn't read a paper_x.*
 - b. *It_x was too dense.*

With a wide-scope-interpretation, the speaker commits to the existence of a referent, which enables subsequent anaphora. Further, Karttunen’s generalization successfully excludes cases where the existence of a referent is not entailed:

- (33) Based on [Karttunen \(1976\)](#): (10b, 11a, 14b, 17)
- a. *Bill can make a kite^{v1}. #It_{v1} has a long string.*
 - b. *John wants to catch a fish^{v2}. #Do you see it_{v2} from here?*
 - c. *John failed to find an answer^{v3}. #It_{v3} was wrong.*
 - d. *I doubt that Mary has a car^{v4}. #Bill has seen it_{v4}.*

Karttunen suggested that anaphora is constrained by semantic inferences about the factuality of embedded contents. While he focused on antecedent contexts, the remainder of this section shows that accessibility is also influenced by the factuality of the anaphor context.

3.2.2 Hypothetical antecedents and local accommodation

Sometimes anaphora is possible even when the existence of a referent isn’t entailed. [Roberts \(1987\)](#) identified *modal subordination*, where hypothetical indefinites provide antecedents for anaphors in non-veridical contexts-

- (34) [Roberts, 1989](#): (11)
- a. *A wolf_x might walk in.*
 - b. *#It_x is gray.*
 - c. *It_x would eat you first.*

Although (34a) does not entail the existence of a referent, anaphora is possible in (34c), with the following paraphrase: “If a wolf were to walk in, that wolf would eat you first.” Importantly, the anaphor is also embedded in a non-veridical context, thus not committing the speaker that *it would eat you first*. A similar situation arises in third-person speech reports:

- (35) Based on [Karttunen \(1976\)](#): (19)
- Bill says he saw a lion_x on the street. He claims it_x had escaped from the Zoo.*

Here, the antecedent *a lion* and the pronoun *it* are interpreted within the non-veridical report of Bill’s claims, again suggesting that hypothetical indefinites can act as antecedents for hypothetical anaphora. Karttunen remarks on such cases that language users “must be sensitive to the semantic properties of verbs that take sentential complements” and “distinguish what exists for the speaker from what exists only for somebody else” (p. 14).

Formal semantics models the beliefs of different epistemic agents by allowing modals and clause-embedding predicates to introduce quantification over possible worlds (based on [Carnap 1947, et seq.](#), e.g., [Kratzer 1977, 1981](#)). These operators specify a set of possible worlds where the embedded content holds (e.g., a wolf does walk in), without necessarily asserting that it actually true. Based on that, the possibility of anaphora in the above cases can be understood in terms of local accommodation ([Heim 1983b](#)): Pronouns presuppose the existence of a referent, and this presupposition can be locally accommodated—assumed to be true in the set of worlds associated with the embedded content. Accordingly, (34) and (35) can be paraphrased as in (36).

- (36) a. A wolf might walk in. It would be the case that (the wolf exists and) it eats you first.
 b. Bill says he saw a lion on the street. He claims (the lion existed and) it had escaped from the Zoo.

The possibility of local accommodation suggests a refined generalization:

- (37) **Generalization about accessibility** (final)
 Individual anaphora is possible just in case the pronoun's presupposition that its referent exists can be accommodated in its local context.

(37) also predicts that factual anaphora should allow for local accommodation, as long as that doesn't create a contradictory discourse. This indeed occurs in third-person attitude ascriptions:

- (38) [Karttunen 1976](#): (18)
Bill doubts that Mary has a car_x. I have seen it_x.

The first sentence reflects a third persons' epistemic state, not committing the speaker that *Mary has a car*. However, when asserting *I have seen it*, they presuppose, and accommodate, that said car exists. We see something similar, when speakers disagree about whether a referent exists:

- (39) Disagreement ([Hofmann 2019](#): 2d)
 A: *There isn't a bathroom_x in this house.*
 B: *(What are you talking about?) It_x's right over there.*

In (39), speaker A's assertion does not commit B to the existence of a bathroom, but B commits to this by using a pronoun that presupposes it. This suggests that any DP can be an antecedent if the speaker using the anaphor commits to the existence of its referent, even if by accommodation.

3.2.3 Revisiting the modal subordination contrast

The possibility of accommodation prompts us to revisit the modal-subordination contrast in (34). At face value, we incorrectly predict that local accommodation should be possible in both (34b+c).

- (34) a. *A wolf_x might walk in.*
 b. *#It_x is gray.*
 c. *It_x would eat you first.*

To understand the unavailability of (34b), we can consider pragmatic reasons why the veridical anaphors are inconsistent with indefinite antecedents under *might*: Possibility claims often carry a scalar implicature that negates the corresponding necessity claim. Scalar implicatures are typically strong inferences and that require explicit cancellation ([Gazdar 1979](#); [Sauerland 2004](#); [Fox 2014](#)). When such cancellation is signaled explicitly, factual anaphora with antecedents under *might* becomes possible:

- (40) *There might be a bathroom in this house. In fact, I just remembered, it's upstairs.*

This explanation aligns with a view that centers discourse consistency and accommodation: the possibility of anaphora depends on both literal truth conditions and contextual inferences, including implicatures, about the invoked referents.

3.2.4 Counterfactual antecedents

Anaphora is also possible with antecedents under negation when the existence of a referent is denied, for instance in cases of modal subordination with negative antecedents (41a) and “negative subordination” (Kibble 1994) where both the antecedent and the anaphor are in the scope of a negation (41b).

- (41) a. *There isn't a bathroom_x in this house. It_x would be easier to find.*
 b. ‘Negative subordination’:
There isn't a bathroom_y in this house, so you won't be able to find it_y.

In both cases, the initial sentence denies the existence of the mentioned bathroom. The existential presupposition of the pronoun is locally accommodated under negation, enabling anaphora. In line with the generalization (37), these cases illustrate that anaphora is possible even with counterfactual antecedents, whenever the existential presupposition is consistently accommodated in a non-veridical or anti-veridical context.

3.3 Veridicality and consistency

Factual antecedents are the most flexible and can support anaphora of any epistemic status. Hypothetical antecedents enable anaphora whenever the existential presupposition can be locally accommodated. Counterfactual antecedents support anaphora only in anti-veridical contexts or semantically non-veridical contexts that remain consistent with an anti-veridical interpretation.

The systematic constraints on anaphora based on entailments about their referents suggest a unified analysis. The observed patterns can be explained based on two key principles of discourse interpretation: First, pronouns carry an existential presupposition, which can be locally accommodated; and second, discourse must avoid contradictions to maintain internal consistency.

The consistency requirement is as crucial as the existential presupposition. The veridicality-constraints apply only when the discourse segments containing the antecedent and the one containing the anaphor are interpreted in conjunction. For example, discourse relations like disjunction (4b) or disagreement (39) do allow for unembedded pronouns with hypothetical or counterfactual antecedents:

- (4b) Bathroom-disjunctions
Either there isn't a bathroom_x in this house or it_x's in a funny place.
- (39) Disagreement
 A: *There isn't a bathroom_x in this house.*
 B: *(What are you talking about?) It_x's right over there.*

These apparent counterexamples, in fact, align with the above generalizations when understood within their respective non-veridical discourse contexts. Both disjunction and disagreement are characterized as *non-veridical discourse relations* in Segmented DRT (Asher 1993; Asher and

Lascarides 2003) which don't require non-contradiction between the discourse segments to form a consistent discourse. Thus, the existential presupposition of the anaphors in (41) can also be understood as being locally accommodated, in terms of the generalization (37).

4 Analyses of intensionally unspecific anaphora

This section asks what kind of semantic machinery is needed when anaphoric antecedents do not denote any actual individual. Section 4.1 highlights the problem that the lack of a real-world denotations pose for a referential characterization of anaphora. Further, the section uses intensionally unspecific anaphora to continue comparing descriptive and dynamic approaches to anaphora: While descriptive approaches can explain these anaphors by assuming they introduce an existential presupposition which can be locally accommodated in hypothetical and counterfactual contexts, dynamic accounts need to further generalize the notion of discourse reference to include intensionally unspecific cases. Section 4.2 will briefly motivate extending dynamic approaches in this way. The rest of the section evaluates dynamic accounts of intensionally unspecific anaphora by establishing four example cases as empirical checkpoints for their predictions: double negation, bathroom disjunctions, modal subordination, and counterfactual anaphora. We first test a dynamic account that relativizes variable denotations to epistemic possibilities in the context set (Section 4.3), then refine it by allowing variables to denote individual concepts, thus separating referential from epistemic information and making room for counterfactual anaphora (Section 4.4). Finally, Section 4.5 summarizes the comparative findings.

4.1 The reference problem and a descriptive solution

Section 4.1 highlighted that quantificationally unspecific anaphora challenge a referential characterization of anaphora. While the descriptive approach does not treat coreference as essential, the dynamic approach broadens the concept of (discourse) reference to include anaphoric coreference relative to quantificational instances. Intensionally unspecific anaphora add a further complication, lacking real-world referents even in a relative sense. This conflicts with traditional, externalist views of reference, as a link between linguistic expressions and real-world entities external to the speaker's mind (Putnam (1975) *et seq.*; see Gross 2015 for an overview).

The descriptive response offers an analysis of hypothetical antecedents that is compatible with an externalist view (van Rooij 2001; Van Rooij 2006; Keshet and Abney 2024). For instance, the simple descriptive system from Section 2.3 derives anaphora with doubly negated antecedents:

- (42) a. *It's not true that there is no bathroom.* $\rightsquigarrow \neg\neg\exists x(\text{bathroom}(x))$
 b. *It's upstairs* $\rightsquigarrow \text{upstairs}(\iota P)$, $P = \lambda x.(\text{bathroom}(x))$

(42a) and (42b) are both true when exactly one bathroom is located upstairs, predicting that anaphora is possible. For bathroom-disjunctions (43), the existential presupposition is locally accommodated in the context of the second disjunct, rendering the disjunction true in situations where no bathroom exists or a unique bathroom is located upstairs.

- (43) a. *Either there is no bathroom, or it's upstairs.* $\rightsquigarrow (\neg\exists x(\text{bathroom}(x)) \vee \text{upstairs}(\iota P))$
 b. $P = \lambda x.(\text{bathroom}(x))$

Offering a presuppositional explanation of accessibility, the descriptive approach derives that anaphora is possible when the pronominal presupposition can be consistently accommodated. In contrast, the dynamic approach needs to further expand its notion of reference to include drefs for entities that may not exist in the actual world, aligning more with internalist views of reference.⁵ At this point, the descriptive approach appears more successful: it aligns with traditional notions of reference and naturally extends to intensionally unspecific anaphora via a presuppositional account of accessibility, avoiding failures of classic dynamic approaches, which incorrectly rule out hypothetical and counterfactual antecedents.

4.2 Dynamic analyses—why bother?

Given the successes of the descriptive approach, one might question the need for dynamic analyses. Critics of the descriptive approach often highlight its assumption of a uniqueness presupposition, which is problematic in cases like Heim’s sage plant example:

(44) Heim 1982: (2:12)

Everybody who bought a sage plant here bought eight others along with it.

(44) is possible apparently without a unique referent, which the descriptive approach rules out (Heim 1982, 1990, though see King and Lewis 2021 for possible resolutions; and van Rooij 2001 arguing that pronouns are generally interpreted uniquely/exhaustively).

Beyond truth-conditions, a key advantage of dynamic semantics is that treating anaphora as discourse variables naturally extends to other domains, including temporal anaphora (Partee 1973, 1984; Kamp 1981a; Stone 1997) and propositional ones (e.g., Hankamer and Sag 1976; Asher 1993; Snider 2017). Anaphora in these domains are harder to account for descriptively, illustrated here for temporal anaphora.

Partee (1973) distinguished three ways that pronominal denotations are determined: Besides bound and anaphoric uses, pronouns can be interpreted deictically, referring to a specific individual in the extralinguistic context. She observes that tenses can also be interpreted in these three ways, illustrated for the future tense in (45):

(45) Based on Partee 1973: (14-16)

- a. Deictic: *John will leave immediately.*
- b. Anaphoric: *Susan is about to come in. John will leave immediately.*
- c. Bound: *Whenever Susan comes in, John immediately leaves.*

In (45a), the time of John leaving is identifiable based on the speech situation. In (45b), the tense anaphorically depends on the previous clause (i.e., John will leave immediately after Susan coming in). (45c) suggests multiple instances of John leaving, each in the future of Susan coming in. Here, the interpretation depends on the quantifier *whenever*, similar to bound anaphora in (2a).

However, Partee also identifies a major difference: while non-deictic individual pronouns require an overt antecedent (e.g., a name or a quantifier), non-deictic tenses can occur with or without explicit temporal modifiers. Because tense is an inflectional category, it is unclear what a “tempo-

⁵In internalist semantic theories (see Riemer 2015 for an overview), denotations are treated as concepts in the mind of the language user, rather than as external objects (e.g., Chomsky 2000; Jackendoff 2002; Pietroski 2017).

ral definite description” would even look like, or how to extend a descriptive account to tense in a principled way. Analyzing temporal reference with discourse variables effectively explains the parallels (Kamp 1981a; Partee 1984), offering a strong motivation for extending dynamic semantics to provide a general theory of anaphora, including intensionally unspecific cases.

4.3 Relativizing drefs to possible worlds

To extend the notion of discourse referents to intensionally unspecific anaphora, we allow drefs to reference different objects across possible worlds. One way is treating context sets as sets of possibilities, each represented by an assignment function mapping variables relative to a given world. This can be implemented as a set of world-assignment-pairs (Heim 1982) or by including a designated world variable in the domain of each assignment function (Dekker 1993). Dynamic analyses of hypothetical anaphora then need a mechanism whereby indefinites introduce drefs only in possibilities where their referent exists. This has been applied to modal subordination (Van Rooij 2005; Brasoveanu 2006), as well as doubly negated antecedents and bathroom anaphora (Gotham 2019; Elliott 2020, 2022; Mandelkern 2022). Here, I illustrate this approach by allowing drefs to be introduced using a projective inference associated with indefinites (c.f., Gotham 2019; Mandelkern 2022). This means that indefinites always introduce a dref, even when embedded under non-veridical operators. Here, this is implemented within a version of File Change Semantics (FCS, Heim 1983a, see also Elliott 2020, 2022), where contexts are sets of world-assignment pairs (definitions in Appendix C).

4.3.1 The basic system

Variable update. We adapt the DPL system from Section 2.4 to work with contexts containing world-assignment pairs, starting with variable update:

$$(46) \quad \text{FCS variable update: } \llbracket [x] \rrbracket^{M,C} = \{ \langle w, h \rangle \mid \langle w, g \rangle \in C, \exists d \in D : h = g[x \mapsto d] \}$$

Given a model M and input context C , (46) defines an output context, containing world-assignment pairs where a dref x is randomly introduced to the assignments. For illustration, consider a model M_2 , where: $D_w = \{w_1, w_2, w_3, w_4\}$ is the domain of worlds, and $D_e = \{c_1, c_2, \text{mary}\}$ is the domain of individuals. An initial context $C_0 = \{ \langle w_1, g \rangle, \langle w_2, g \rangle, \langle w_3, g \rangle, \langle w_4, g \rangle \}$, where g is an arbitrary assignment can be updated with (46), as shown in Figure 5.

Each element of the context represents an epistemic possibility, where the worlds are candidates for the actual world. Updates can introduce new variable mappings or eliminate possibilities.

Predication. To interpret predication intensionally, (47) outputs all world-assignment pairs $\langle w, g \rangle \in C$ where $g(x)$ is a car in w . This update is a test, containing only unchanged elements from the input.

$$(47) \quad \text{FCS predication: } \llbracket [\text{car}(x)] \rrbracket^{M,C} = \{ \langle w, g \rangle \in C \mid g(x) \in I_M(\text{car})(w) \}$$

Sentences with indefinites are represented as in (48) (full derivations in Appendix C).

$$(48) \quad \text{a. } \textit{Mary owns a car}_x.$$

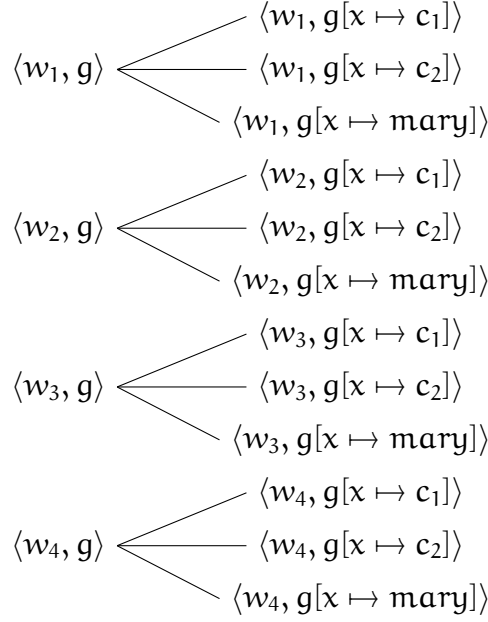


Figure 5: FCS variable update

- b. $\llbracket [x]; [\text{car}(x)]; [\text{own}(m, x)] \rrbracket^{M, C} =$
 $\{ \langle w, g[x \mapsto d] \rangle \mid \langle w, g \rangle \in C, d \in I_M(\text{car})(w), \langle I_M(m), d \rangle \in I_M(\text{own})(w) \}$

(48) defines an output containing all pairs $\langle w, g \rangle \in C$ where $g(x)$ is a car owned by Mary in w . Assume for M_2 that c_1, c_2 are cars in all worlds, Mary owns c_1 in w_1 , and c_2 in w_2 . Figure 6 illustrates the update of C_0 with (48) given M_2 .

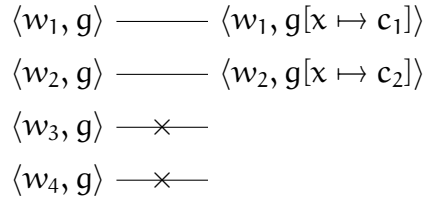


Figure 6: Update with *Mary owns a car*.

Discourse update and truth. In this dynamic system, utterance meaning is characterized by the information added to the discourse. Because the context stores epistemic information about the candidates for the actual world, truth is understood relative to the discourse context:

- (49) a. Discourse update: $C + S := \llbracket S \rrbracket^{M, C}$
b. Truth: S is true relative to model M and context C , iff $\llbracket S \rrbracket^{M, C} \neq \emptyset$

An update is true iff it has a non-empty set of output possibilities.

Anaphora. Pronouns are discourse variables:

- (50) a. $\underline{It}_x$'s red.
 b. $\llbracket \text{red}(x) \rrbracket^{M,C} = \{\langle w, g \rangle \in C \mid g(x) \in I_M(\text{red})(w)\}$

For instance, if c_1 is red and c_2 is not, applying (50) to the context in Figure 6 retains only the possibility $\langle w_1, g[x \mapsto c_1] \rangle$ where $g(x)$ is red in w (see Figure 7).

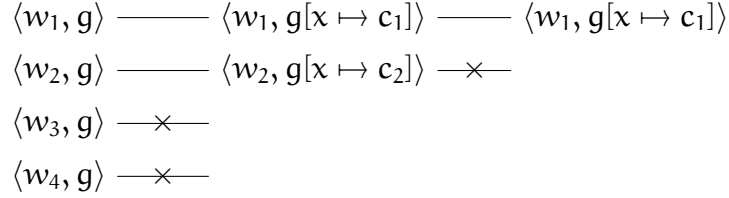


Figure 7: Update with *Mary owns a car. It is red.*

4.3.2 Negation and doubly negated antecedents

Negation. In their at-issue content, negated updates are still tests that eliminate possibilities where the prejacent is true.

- (51) FCS negation: $\llbracket \neg \phi \rrbracket^{M,C} = \{\langle w, g \rangle \in C \mid \llbracket \phi \rrbracket^{M, \{\langle w, g \rangle\}} = \emptyset\}$

Accordingly, an update involving a counterfactual indefinite (52) retains all elements $\langle w, g \rangle \in C_0$ where Mary owns no car in w , without updating g (illustrated for C_0, M_2 in Figure 8).

- (52) a. *Mary doesn't own a car_x*
 b. $C_0 + \neg([x]; [\text{car}(x)]; [\text{own}(m, x)]) =$
 $\{\langle w, g \rangle \in C_0 \mid \llbracket ([x]; [\text{car}(x)]; [\text{own}(m, x)]) \rrbracket^{M, \{\langle w, g \rangle\}} = \emptyset\}$

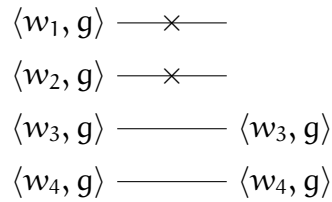


Figure 8: Update with *Mary doesn't own a car.*

Doubly negated indefinites (53), similarly, don't (yet) introduce drefs.

- (53) a. It's not the case that Mary doesn't own a car_x.
 b. $C_0 + \neg \neg([x]; [\text{car}(x)]; [\text{own}(m, x)]) =$
 $\{\langle w, g \rangle \in C_0 \mid \llbracket ([x]; [\text{car}(x)]; [\text{own}(m, x)]) \rrbracket^{M, \{\langle w, g \rangle\}} \neq \emptyset\}$

Because negation is externally static, a double negation update retains input elements where the prejacent is true without updating them (Figure 9).

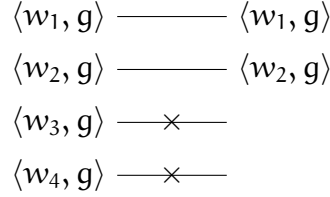


Figure 9: Update with *It's not the case that Mary doesn't own a car*.

Variable update via a projective inference. We let indefinites introduce drefs in each possibility where a referent exists (Van Rooij 2005; Brasoveanu 2006; Gotham 2019; Elliott 2020, 2022; Mandelkern 2022). To model cases where a variable has no referent, we include a dummy individual # in the domain of individuals, acting as a universal falsifier making any lexical relation false (following Brasoveanu 2006). Indefinites introduce a dref in all possibilities with an existing referent through a projective inference (cf. Gotham 2019; Mandelkern 2022):

- (54) Indefinites: $\exists x(\phi)$
- a. Assertion: $[x]; \phi$
 - b. Projective inference: $[x]; (\neg\phi \leftrightarrow [x = \#_e])$

The projective inference (54b) introduces a dref for x , and states that x maps to # iff ϕ is false, or to an individual otherwise. The biconditional is defined as (55), retaining all elements of the input where both conjuncts are true (non-empty) or false.

$$(55) \quad \llbracket [\phi \leftrightarrow \psi] \rrbracket^{M,C} = \{ \langle w, g \rangle \in C \mid \llbracket \phi \rrbracket^{M, \{ \langle w, g \rangle \}} = \llbracket \psi \rrbracket^{M, \{ \langle w, g \rangle \}} \}$$

Factual indefinites, which entail that their referent exists, introduce a dref throughout the entire context set—making it *familiar*, in Heim's (1983a) terms. For double negation, we derive this as:

- (56) a. *It's not the case that Mary doesn't own a car*_x. $\rightsquigarrow \neg\neg\exists x([car(x)]; [own(m, x)])$
- b. Projective inference: $[x]; ((\neg([car(x)]; [own(m, x)])) \leftrightarrow [x = \#_e])$
 - c. Assertion: $\llbracket \neg\neg([x]; [car(x)]; [own(m, x)]) \rrbracket^{M,C} = \{ \langle w, g \rangle \in C \mid \llbracket ([x]; [car(x)]; [own(m, x)]) \rrbracket^{M, \{ \langle w, g \rangle \}} \neq \emptyset \}$

Using an updated model $M_2^\#$ (like M_2 , but including # in the domain of individuals), and input C_0 , the sequence of updates is illustrated in Figure 10. By assumption, we first apply the projective inference (56b): For each element $\langle w, g \rangle \in C_0$, if Mary owns a car in w , a dref x for that car is introduced; otherwise, $g(x)$ is the dummy #. Then, (56a) eliminates possibilities where the prejacent is false, resulting in the same output as a simple positive assertion (48), explaining why it allows for a subsequent anaphor in the same way.

4.3.3 Modal subordination and partially familiar drefs

Epistemic possibility. To analyze modal subordination using this projective inference, we adopt Veltman's (1996) simple test semantics for the epistemic possibility modal *might*:

$$(57) \quad \text{FCS epistemic possibility: } \llbracket [\Diamond\phi] \rrbracket^{M,C} = \{ \langle w, g \rangle \in C \mid \llbracket \phi \rrbracket^{M,C} \neq \emptyset \}$$

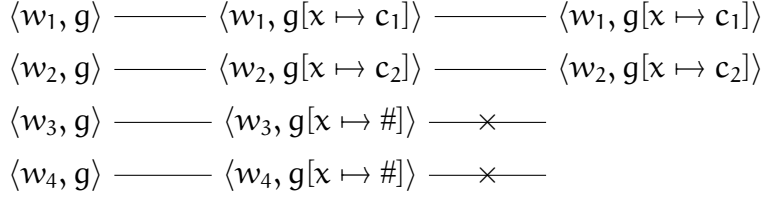


Figure 10: Update with *It's not the case that Mary doesn't own a car*.

Updates with *might* retain the full input set iff some possibility makes the prejacent true. For a modalized indefinite (58), the assertion (58c) is interpreted after the projective inference (58b).

- (58) a. *Mary might own a car*_x. $\rightsquigarrow \Diamond \exists x([car(x)]; [own(m, x)])$
 b. Projective inference: $[x]; ((\neg([car(x)]; [own(m, x)]) \leftrightarrow [x = \#_e])$
 c. Assertion: $\Diamond([x]; [car(x)]; [own(m, x)])$

After (58b) is applied, some possibilities include a dref x for Mary's car, while those possibilities where no such object can be found map x to $\#$ (Figure 11). Because Mary owns a car in some possibilities, the modal assertion (58c) retains the entire context set. This allows hypothetical indefinites to introduce *partially familiar* drefs, which are defined in a subset of the possibilities in the context (terminology from Elliott 2020, 2022).

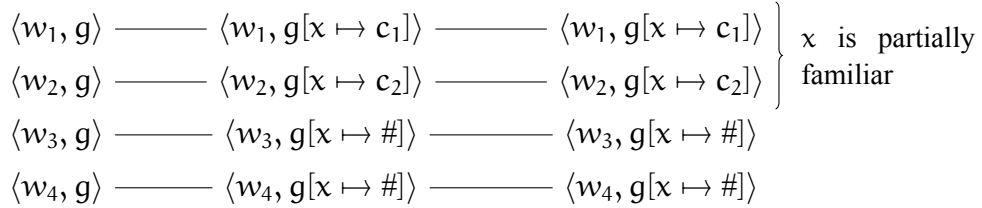


Figure 11: Update with *Mary might own a car*, making a dref partially familiar.

Remote epistemic necessity. To give a suitable semantics for *would*, let's consider which possibilities should be eliminated when updating with (59).

- (59) *It_x would be red.*

For $M_2^\#$, suppose that Mary likes only red things in worlds w_1 and w_3 , making it inconceivable for her to have a non-red car. We assume an epistemic accessibility relation R , which, for each possible world w specifies a set of worlds $R(w)$ that represent epistemic possibilities based on what is known in w (Kratzer 1981). Here, we let $R(w_1) = R(w_3) = \{w_1, w_3, w_4\}$, so the worlds accessible from w_1, w_3 , are those where Mary doesn't own a non-red car. Similarly, suppose that in w_2 and w_4 , Mary likes only non-red things, making it inconceivable for her to have a red car, and $R(w_2) = R(w_4) = \{w_2, w_3, w_4\}$. The update with (59) should leave open whether Mary actually owns a car, but eliminate worlds where she only has a non-red car (w_2), or worlds where a non-red car is conceivable but a red car is not (w_4), illustrated in Figure 12.

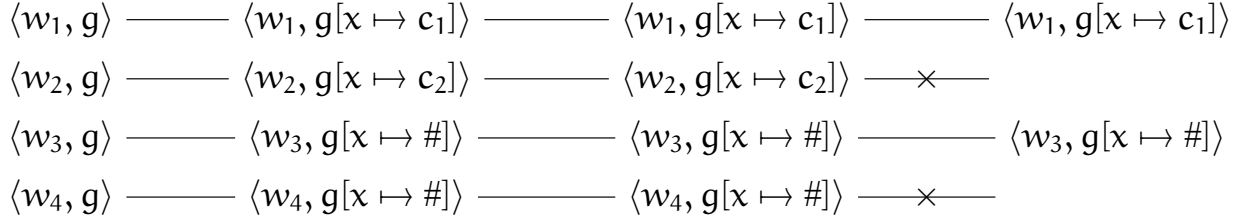


Figure 12: Update with *Mary might own a car. It would be red.*

We derive this update by analyzing *would* as a remote epistemic necessity modal. Following Kratzer 1977, 1981, 1991, this analysis builds on the analysis for *necessity modals* (60), which require that every world in the modal restrictor set makes the prejacent true. For *epistemic modals*, the restrictor is determined by an epistemic accessibility relation (cf. Kratzer’s *modal base*).⁶ Accordingly, an update with a (non-remote) epistemic necessity modal (60) retains all possibilities $\langle w, g \rangle$, where $g(x)$ is red in all epistemically accessible worlds.

$$(60) \quad \llbracket \Box([\text{red}(x)]) \rrbracket^{M,C} = \{ \langle w, g \rangle \in C \mid \forall w' \in R(w) : \llbracket [\text{red}(x)] \rrbracket^{M, \{ \langle w', g \rangle \}} \neq \emptyset \}$$

This does not yet derive the update in Figure 12. Since R is a reflexive relation ($\forall w : w \in R(w)$), each world includes itself as a possibility. Therefore, (60) eliminates any possibility where $g(x)$ is not red (including $\langle w_3, g[x \mapsto \#] \rangle$). This makes (60) more appropriate to represent a sentence with a (simple) epistemic necessity modal, like *It must be red*. To capture the intended meaning (“If Mary had a car, that car would be red”), we limit the modal restrictor to the “remote” possibilities where Mary actually has a car, defining it as the intersection of $R(w)$ and an additional set of worlds B , which is provided contextually (this question of how it is provided in context is discussed below).

$$(61) \quad \llbracket \text{would}([\text{red}(x)]) \rrbracket^{M,C} = \{ \langle w, g \rangle \in C \mid \forall w' \in (R(w) \cap B) : \llbracket [\text{red}(x)] \rrbracket^{M, \{ \langle w', g \rangle \}} \neq \emptyset \}$$

If we suppose that $B = \{w_1, w_2\}$ (the worlds where Mary has a car), we predict the update shown in Figure 12: The update retains all possibilities $\langle w, g \rangle$, such that each epistemically accessible world where Mary has a car $w' \in (R(w) \cap B)$, is one where $g(x)$ is red in w' .

How to determine the restrictor? Previous literature suggests that the extra restriction B to the modal base is retrieved anaphorically, thereby enabling modal subordination (Roberts 1987, 1989; Frank 1996; Brasoveanu 2006, 2010). While this assumption is motivated by the context-dependent truth-conditions of epistemic modals (Kratzer 1981; Frank 1996; Frank and Kamp 1997; Geurts 1999; Anand and Hacquard 2013), it is not necessary for enabling anaphora, as pronouns in non-veridical contexts very generally allow for hypothetical antecedents.

If it was necessary, we would expect a similar mechanism for other cases of hypothetical anaphora. Indeed, certain clause-embedding predicates have been argued to anaphorically restrict the modal base (Asher 1987; Maier 2015) but anaphora with hypothetical antecedents is even more

⁶The denotation for epistemic possibility modals in (57) fits in this theory of modality (as existential quantifiers over epistemically accessible worlds) if we assume that the context set represents the possibilities that are epistemically accessible from the world of evaluation.

flexible. Recall that anaphora is possible whenever the existential presupposition is consistently accommodated, even in unembedded contexts:

(38) *Bill doubts that Mary has a car_x. I have seen it_x.*

For unembedded factual pronouns (38), it is harder to motivate a contextually provided modal base. Such examples suggest a more general mechanism for retrieving partially familiar drefs. In fact, the FCS system with partially familiar drefs defined here (wrongly) predicts that a veridical anaphor following the update with *might* (62) is possible:

(62) *Mary might own a car_x. # It_x is red.*

The eliminative update mechanism simply removes all possibilities that don't satisfy the stated conditions (see Figure 13), predicting that anaphora is possible.

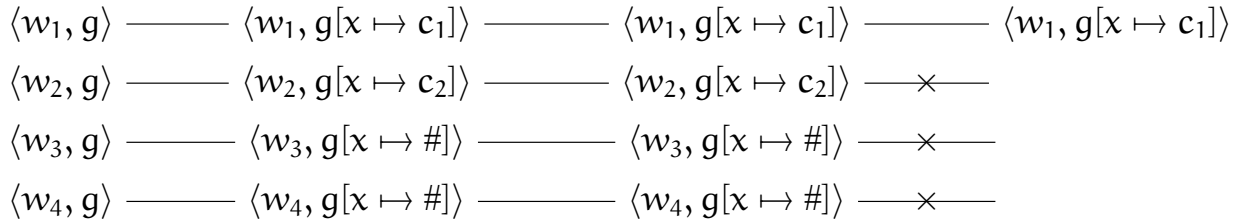


Figure 13: (Predicted) update with *Mary might own a car. It is red.*

To illustrate the issue, consider the update with the veridical anaphor (50), repeated here:

(50) a. *It_x is red.*
b. $\llbracket [\text{red}(x)] \rrbracket^{M,C} = \{ \langle w, g \rangle \in C \mid g(x) \in I_M(\text{red})(w) \}$

(62) can only be true if the variable x is defined, and more generally, any update entails definedness. Therefore, the eliminative update mechanism directly predicts local accommodation of the existential presupposition throughout. However, as discussed in Section 3.2.2, this makes incorrect predictions for (61). To rule out related cases, Mandelkern (2022) proposes that pronouns presuppose that the associated variable is defined throughout the entire input context set. Alternatively, Section 3.2.2 suggested that the unacceptability of (62) may stem from Gricean principles of informativity and resulting inferences about speaker commitments. Under that view, the semantic system could, in principle, derive an update like Figure 13, but it is blocked by pragmatic constraints—though these constraints must be made explicit to make concrete predictions.

4.3.4 Bathroom disjunctions

We state the semantics for disjunction as follows:

(63) FCS disjunction
 $\llbracket [\phi \vee \psi] \rrbracket^{M,C} = \{ \langle w, g \rangle \in C \mid \llbracket [\phi] \rrbracket^{M, \{ \langle w, g \rangle \}} \cup \llbracket [\psi] \rrbracket^{M, \{ \langle w, g \rangle \}} \neq \emptyset \}$

A disjunctive update retains any element of the input context that makes at least one disjunct true. Interpreting the bathroom-disjunction (64) in the context of the projective inference (64b) given $M_2^\#, C_0$ yields the update shown in Figure 14.

- (64) a. Either Mary doesn't own [a car]_x, or it_x's red. $\rightsquigarrow$
 $(\neg \exists x([\text{car}(x)]; [\text{own}(m, x)])) \vee [\text{red}(x)]$
 b. Projective inference: $[x]; ((\neg([\text{car}(x)]; [\text{own}(m, x)])) \leftrightarrow [x = \#_e])$
 c. Assertion: $(\neg([x]; [\text{car}(x)]; [\text{own}(m, x)])) \vee [\text{red}(x)]$

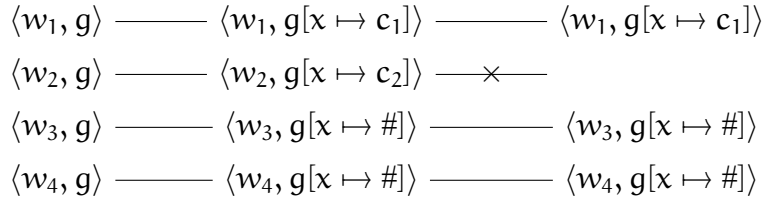


Figure 14: Update with *Either Mary doesn't own a car, or it's red.*

The disjunctive assertion (63c) is a test that retains elements $\langle w, g \rangle$ where: $g(x)$ is a red car that Mary owns in w (e.g., $\langle w_1, g[x \mapsto c_1] \rangle$), or there is no way to make $g(x)$ point to a car that Mary owns in w (as for the other two remaining possibilities).

4.3.5 Discussion and limitations

The approaches using projective content to introduce drefs to possibilities where a referent exists are compatible with an externally static semantics for non-veridical operators. Gotham (2019) integrates projective variable update with classic dynamic semantics, and Mandelkern (2022) develops a fully static semantics for at-issue content, while indefinites and pronouns manipulate context sets via a separate dimension of presupposition-like bounds. Elliott (2020, 2022), instead, integrates positive and negative updates associated with indefinites into a fully dynamic system, deriving relevant generalizations from the truth-functional contributions of sentential operators.

Empirically, however, these approaches make similar predictions for the cases discussed here. Importantly, they share a key limitation: because context sets only retain hypothetical drefs for epistemic possibilities that are still live options in discourse, they cannot handle counterfactual anaphora, as in modal subordination with negated antecedents (65).

- (65) *Mary doesn't own a car. It would be parked outside.*

Because the first sentence entails the non-existence of a car, the resulting update eliminates all possibilities where a dref is introduced. The representation is given in (66):

- (66) a. Mary doesn't own [a car]_x $\rightsquigarrow \neg \exists x([\text{car}(x)]; [\text{own}(m, x)])$
 b. Projective inference: $[x]; ((\neg([\text{car}(x)]; [\text{own}(m, x)])) \leftrightarrow [x = \#_e])$
 c. Assertion: $\neg([x]; [\text{car}(x)]; [\text{own}(m, x)])$

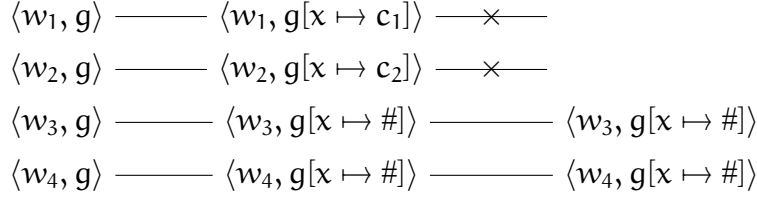


Figure 15: Update with *Mary doesn't own a car*, in the context of the projective inference.

Interpreting the negative assertion (66c) in the context of the projective inference (66b) given M_2, C_0 derives the update illustrated in Figure 15.

This update retains only elements that cannot be updated to assign x to a car that Mary owns. Consequently, any possibility where x is defined and the prejacent is true is eliminated. Thus, (66) fails to make a dref (even partially) familiar, falsely predicting that (65) is unacceptable.

The core problem is that context sets of world-assignment pairs closely link epistemic possibilities (worlds) with referential information (assignments). These discourse states can store referential information only for worlds that remain viable candidates for the actual world. Consequently, once the possibility that Mary owns a car is ruled out, it can no longer be invoked anaphorically, not even hypothetically. By merging epistemic and referential possibilities into a single context set, these accounts inherently rule out counterfactual anaphora, as referents are removed from discourse as soon as their existence is negated.

4.4 Addressing counterfactual anaphora

Counterfactual anaphora show that we need to track referential information for any possible world, even those that are no longer considered epistemically possible. To track hypothetical variable assignments separately from the possible worlds that store epistemic information, we use drefs for *individual concepts*: functions from worlds to individuals (following Carnap 1947; Stone 1999; Stone and Hardt 1999; Hofmann 2019, 2022, see also Aloni 2001; Van Rooij 2006).

4.4.1 Drefs for individual concepts

Variable update is still random, but now the value of each dref is an individual concept—a function from D_w to D_e , mapping each world to an individual. If a dref has no referent in some world, it maps to the dummy individual $\#$. In model $M_2^\#$, where: $D_w = \{w_1, w_2, w_3, w_4\}$ is the domain of worlds, and $D_e = \{\text{mary}, c_1, c_2, \#\}$ is the domain of individuals, there are $|D_e|^{|D_w|} = 4^4 = 256$ such possible functions. Figure 16 illustrates some of them.

$$\begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_1 \\ w_3 \mapsto c_1 \\ w_4 \mapsto c_1 \end{bmatrix} \quad \begin{bmatrix} w_1 \mapsto c_2 \\ w_2 \mapsto c_1 \\ w_3 \mapsto c_1 \\ w_4 \mapsto c_1 \end{bmatrix} \quad \dots \quad \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_2 \\ w_3 \mapsto \text{mary} \\ w_4 \mapsto \# \end{bmatrix} \quad \dots \quad \begin{bmatrix} w_1 \mapsto \# \\ w_2 \mapsto \# \\ w_3 \mapsto \# \\ w_4 \mapsto \# \end{bmatrix}$$

Figure 16: Some possible functions $D_w \mapsto D_e$

4.4.2 The basic system

Here, we use a simplified version of Hofmann’s (2022) Intensional Compositional DRT, which we call *Intensional Logic of Change* (ILC, full definitions in Appendix D). Representations of simple existential updates remain as above:

$$(67) \quad \text{Mary owns } \underline{a} \text{ car}_x \rightsquigarrow ([x]; [\text{car}(x)]; [\text{own}(m)(x)])$$

We do adjust how contexts are structured and how updates apply: a discourse context C is a set of pairs $\{\langle W, g \rangle\}$, where $W \subseteq D_w$ is a set of epistemically possible worlds that are still in play, and g is a variable assignment. ILC formulas are evaluated relative to a model M , a set of worlds W' , and an assignment g . ILC updates (68) apply distributively to an input context C_{in} .

$$(68) \quad \text{ILC discourse update: } C_{in} +^M S := \bigcup_{\langle W, g \rangle \in C_{in}} \{\langle W \cap W', h \rangle \mid h \in \llbracket S \rrbracket^{M, W', g}\}$$

(68) checks, for each pair $\langle W, g \rangle \in C_{in}$, whether it should be updated or eliminated. Output set elements are derived by intersecting W with a set W' that makes the sentence true (i.e. it has a non-empty set of output assignments H for g), and pairing it with one of the output assignments h for W' (i.e., $h \in H$). Based on the sum-operation, the output contains possibilities for all possible choices of $\langle W, g \rangle \in C_{in}$, that satisfy these conditions.

I will explain how an update with (67) is derived after illustrating its outcome for $M_2^\#$, and an input context $C_0 = \{\langle W, g \rangle\}$, where $W = D_w$ and g is an arbitrary assignment (Figure 17).

$$C_0 \longrightarrow \left\langle \{w_1, w_2\}, g[x \mapsto \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_2 \\ w_3 \mapsto \# \\ w_4 \mapsto \# \end{bmatrix}] \right\rangle$$

Figure 17: Update with *Mary owns a car*

The output is a singleton set containing the pair $\langle W_{car}, g' \rangle$, where $W_{car} = \{w_1, w_2\}$ contains the worlds where Mary has a car, and $g'(x)$ is an individual concept mapping to a car in W_{car} and $\#$ elsewhere. This is derived based on intensional versions of variable update and predication.

Variable update. ILC variable update (69) introduces drefs for individual concepts *relative* to a set of worlds W (Stone 1999; Stone and Hardt 1999; Brasoveanu 2006, 2010; Hofmann 2019, 2022).

$$(69) \quad \text{ILC relative variable update} \\ \llbracket [x] \rrbracket^{M, W, g} = \{h \mid \exists f: D_w \mapsto D_e, h = g[x \mapsto f], \forall w \in D_w: w \in W \leftrightarrow x(w) \neq \#\}$$

(69) defines a set of assignments h , where h is an update of g with x , which now stores an individual concept f . For each w in W , $h(x)(w)$ is a genuine individual; for any world w outside W , $h(x)(w)$ maps to $\#$. This ensures that the individual concept associated with a new dref x stores a referent for only the worlds in W , as illustrated for some choices of W in Figure 18.

$$\begin{array}{lcl}
\text{for } W = D_w: & \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_1 \\ w_3 \mapsto c_1 \\ w_4 \mapsto c_1 \end{bmatrix} & \begin{bmatrix} w_1 \mapsto c_2 \\ w_2 \mapsto c_1 \\ w_3 \mapsto c_1 \\ w_4 \mapsto c_1 \end{bmatrix} \quad \dots \quad \begin{bmatrix} w_1 \mapsto \text{mary} \\ w_2 \mapsto \text{mary} \\ w_3 \mapsto \text{mary} \\ w_4 \mapsto \text{mary} \end{bmatrix} \\
\text{for } W = \{w_1, w_2, w_3\}: & \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_1 \\ w_3 \mapsto c_1 \\ w_4 \mapsto \# \end{bmatrix} & \begin{bmatrix} w_1 \mapsto c_2 \\ w_2 \mapsto c_1 \\ w_3 \mapsto c_1 \\ w_4 \mapsto \# \end{bmatrix} \quad \dots \quad \begin{bmatrix} w_1 \mapsto \text{mary} \\ w_2 \mapsto \text{mary} \\ w_3 \mapsto \text{mary} \\ w_4 \mapsto \# \end{bmatrix} \\
\dots & & \\
\text{for } W = \emptyset: & \begin{bmatrix} w_1 \mapsto \# \\ w_2 \mapsto \# \\ w_3 \mapsto \# \\ w_4 \mapsto \# \end{bmatrix} &
\end{array}$$

Figure 18: Individual concepts randomly introduced relative to W

Predication. Predicative updates (70) eliminate possibilities not satisfying the given conditions.

$$(70) \quad \text{ILC predication: } \llbracket [\text{car}(x)] \rrbracket^{M, W, g} = \{h \mid h = g, \forall w \in W: h(x)(w) \in I_M(\text{car})(w)\}$$

(70) retains only elements $\langle W, g \rangle$ for which $g(x)(w)$ is a car for each world w in W .⁷

Truth and choosing the evaluation set. Combining (69) and (70) allows us to derive (71).

$$\begin{aligned}
(71) \quad & \llbracket ([x]; [\text{car}(x)]; [\text{own}(m)(x)]) \rrbracket^{M, W, g} = \\
& \{g[x \mapsto f] \mid f: D_w \mapsto D_e, \forall w \in D_w: w \in W \leftrightarrow x(w) \neq \#, \\
& \forall w \in W: f(w) \in I_M(\text{car})(w), \\
& \forall w \in W: \langle I_M(m), k(x)(w) \rangle \in I_M(\text{own})(w)\}
\end{aligned}$$

As f must designate a car that Mary owns in each $w \in W$, how we choose the epistemic context set W is important. If W contains worlds where Mary has no car, the predication is false, yielding an empty output. It yields a non-empty output for any $W \subseteq W_{\text{car}}$, paired with the corresponding mapping for x . However, although any subset of W_{car} satisfies the predication, the condition is vacuously true for $W = \emptyset$. We exclude this option in discourse by assuming non-empty context sets to ensure that discourses remain consistent and non-contradictory. We say that an update is true iff it yields at least one viable assignment g , paired with a non-empty context set.

$$(72) \quad \text{Truth: } S \text{ is true for } M, C \text{ iff } \exists \langle W, g \rangle \in C +^M S : W \neq \emptyset$$

The update in Figure 17 is derived by assuming exactly $W = W_{\text{car}}$, i.e., the largest set that yields a non-empty output for $\llbracket ([x]; [\text{car}(x)]; [\text{own}(m)(x)]) \rrbracket^{M^\#, W, g}$. Although the semantic system also outputs combinations for other subsets of W_{car} , we typically select the largest context set W compatible with the update. This preference can be motivated by Gricean quantity (assuming that had the speaker intended to commit to something more informative, they would have explicitly done

⁷Including universal quantification over worlds in lexical meanings is the key feature of encapsulated quantification accounts of dependencies of individual anaphora on modal quantification (Stone 1997, 1999; Bittner 2001, 2007).

so; Grice 1975). Another possible explanation is the information-theoretic principle of maximum entropy (the knowledge state that best captures the available information is the one with the most uncertainty; Jaynes 1957). In the following discussion, we assume this pragmatic maximization of context sets and illustrate updates only for the largest possible sets.

Anaphora. The update with an anaphoric pronoun is represented as in (73). Assuming that c_1 is red in all worlds and c_2 is not, the update is illustrated in Figure 19.

- (73) a. $\underline{It}_x$ is red.
 b. $\llbracket [\text{red}(x)] \rrbracket^{M_2^{\#}, W', g} = \{h \mid h = g, \forall w \in W': h(x)(w) \in I_{M_2^{\#}}(\text{red})(w)\}$

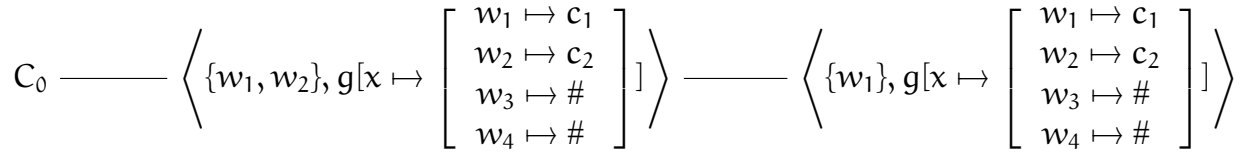


Figure 19: Update with *Mary has a car. It is red.*

We select the largest set of worlds W' making the update true and intersect it with the epistemic context sets from the input. In this example, $W' = \{w_1\}$, where x points to the red car c_1 . Importantly, W' must be a subset of the worlds where the antecedent was introduced, i.e. $W' \subseteq \{w_1, w_2\}$. This “subset condition” arises because $\#$ falsifies any predication, so a world can only be included in W' if x is defined there, i.e., $h(x)(w) \neq \#$ for each $w \in W'$. This derives our generalization from Section 3: an anaphoric pronoun can be interpreted only if the antecedent referent exists in every world of the pronoun’s local context. The possibility of freely choosing the largest W' that satisfies the update predicts a liberal possibility of local accommodation.

4.4.3 Counterfactual anaphora

Indefinites under negation (74) introduce a *dref* relative to the worlds in its local context.

- (74) *Mary doesn't own a car_x.*

Negation introduces a sets of worlds W_2 , which is the local context where the embedded content is true,⁸ and states that the matrix context of evaluation W_1 is the complement of W_2 .

- (75) ILC negation: $\llbracket [\neg\phi] \rrbracket^{M, W_1, g} = \{h \mid \exists W_2 \subseteq D_w : W_2 = \overline{W_1}, h \in \llbracket \phi \rrbracket^{M, W_2, g}\}$

(76) updates the the input context C_0 by introducing a *dref* x for Mary’s car relative to W_2 , the largest set of worlds in which Mary owns a car (i.e., $\{w_1, w_2\}$, see Figure 20).

- (76) a. *Mary doesn't own a car*

⁸We also implicitly assume that the local context is *the largest set* making the prejacent true. While this is not explicitly implemented here, it is necessary to derive the truth-conditions for negation and other propositional operators. For a detailed discussion and formalization of maximization for propositions, see Hofmann 2022.

- b. $\llbracket \neg([x]; [\text{car}(x)]; [\text{own}(m)(x)]) \rrbracket^{M, W, g} =$
 $\{g[x \mapsto f] \mid \exists W' \subseteq D_w : W' = \overline{W},$
 $f: D_w \mapsto D_e, \forall w \in D_w : w \in W' \leftrightarrow x(w) \neq \#,$
 $\forall w \in W' : f(w) \in I_M(\text{car}),$
 $\forall w \in W' : \langle I_M(m), k(x)(w) \rangle \in I_M(\text{own})(w) \}$

$$C_0 \longrightarrow \left\langle \{w_3, w_4\}, g[x \mapsto \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_2 \\ w_3 \mapsto \# \\ w_4 \mapsto \# \end{bmatrix}] \right\rangle$$

Figure 20: Update with *Mary doesn't own a car*.

This update introduces the same variable mappings as the affirmative update above. However, the matrix evaluation context W_1 is the complement of W_2 (i.e., $\{w_3, w_4\}$), restricting the epistemic context sets to contain only W_1 -worlds, where x points to $\#$. This discourse state entails that the referent of x does not exist (77), making it a *counterfactual dref*.

- (77) A dref x is *counterfactual* in C , iff $\forall \langle W, g \rangle \in C (\forall w \in W (g(x)(w) = \#))$.

The epistemic status of this dref differs from the affirmative case (4.4.2), where the existence of the referent is entailed, because the discourse contains only epistemic possibilities where x refers to an actual entity (not $\#$). Here, x is a *factual dref* (78).

- (78) A dref x is *factual* in C , iff $\forall \langle W, g \rangle \in C (\forall w \in W (g(x)(w) \neq \#))$, and hypothetical otherwise.

We can now explain why a counterfactual dref cannot be picked up by a factual anaphor (79).

- (73) a. *It_x is red.*
b. $\llbracket [\text{red}(x)] \rrbracket^{M, W, g} = \{h \mid h = g, \forall w \in W : h(x)(w) \in I_M(\text{red})(w)\}$

To update the context in Figure 20 with (73), we choose $W = \{w_1\}$, the largest set of worlds where x is defined and red. However, the epistemic sets in the discourse are already restricted to $\{w_3, w_4\}$, where x is not defined. This contradiction renders the update impossible, making a factual anaphor unacceptable in this context.

In contrast, an anaphor under *would* is possible (79a). We adapt the semantics for *would* from section Section 4.3.3 to ILC by letting it introduce a local context W_2 as the largest set of worlds where the prejacent is true. The modal relation between W_2 and the matrix context W_1 is expressed as follows: for each $w \in W_1$, the intersection of worlds accessible from w and a contextually determined set B contains only W_2 -worlds $((R(w) \cap B) \subseteq W_2)$. Because this relation does not require that W_2 is entailed by W_1 , it is non-veridical (Figure 21).

- (79) a. *It_x would be red.*
b. $\llbracket [\text{would}([\text{red}(x)])] \rrbracket^{M, W_1, g} =$
 $\{h \mid \exists W_2 \subseteq D_w : h \in \llbracket [\text{red}(x)] \rrbracket^{M, W_2, g}, \forall w \in W_1 : (R(w) \cap B) \subseteq W_2\}$

To interpret (79b), we choose a suitable W_1 , W_2 , and B . As in Section 4.3.3, we assume that $B = \{w_1, w_2\}$, the worlds where Mary has a car. W_2 contains all worlds where x is red (i.e., $\{w_1\}$), and W_1 contains all worlds where the remote modal restrictor $B \cap R(w)$ only contains W_2 -worlds. For $M_2^\#$, recall the following epistemic accessibility relation: $R(w_1) = R(w_3) = \{w_1, w_3, w_4\}$; $R(w_2) = R(w_4) = \{w_2, w_3, w_4\}$. Therefore, we choose $W_1 = w_1, w_3$, where the accessible worlds where Mary has a car (w_1) are also worlds where x is red.

$$C_0 \longrightarrow \left\langle \{w_3, w_4\}, g[x \mapsto \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_2 \\ w_3 \mapsto \# \\ w_4 \mapsto \# \end{bmatrix}] \right\rangle \longrightarrow \left\langle \{w_3\}, g[x \mapsto \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_2 \\ w_3 \mapsto \# \\ w_4 \mapsto \# \end{bmatrix}] \right\rangle$$

Figure 21: Update with *Mary doesn't own a car. It would be red.*

The resulting output contains pairs $\langle W', h \rangle$, s.t. $\langle W, g \rangle$ was in the input, the epistemic sets are narrowed by the new information ($W' = W \cap W_1$), and paired with an assignment h resulting from (79b). Since the update has a non-empty output, it is predicted to be acceptable here.

ILC discourse states track which drefs exist in which worlds, and which worlds are still epistemically possible. Updating epistemic and referential information somewhat independently allows for analyzing counterfactual anaphora, which systems like in Section 4.3 cannot. Despite this dissociation, storing epistemic information about drefs is crucial. Variables refer to concepts that may be factual, hypothetical, or counterfactual. This epistemic determines dref accessibility: because variables are interpreted intensionally, relative to their local context, updates with variables can only be true when their referent exists locally. The flexibility in choosing local contexts derives a liberal possibility for accommodating the existence presupposition.

4.5 Interim conclusion

Intensionally unspecific anaphora—where the antecedent does not refer to an actual, real-world entity—challenge traditional referential characterizations. Descriptive approaches handle such examples by presupposing the existence of a referent and allowing local accommodation. Although debates persist about whether pronouns presuppose uniqueness, treating them as definites with an existential presupposition naturally explains why anaphora is possible whenever that presupposition can be satisfied or accommodated.

Dynamic accounts need extra machinery to capture the same generalization. This section showed that interpreting discourse variables intensionally—separating referential and epistemic information—allows dynamic semantics to handle intensionally unspecific anaphora. Although the descriptive approach initially appears simpler, dynamic semantics addresses a wider range of phenomena (including temporal and propositional anaphora) by treating pronouns as discourse variables. This provides strong motivation for adopting a variable-based analysis of pronouns, despite the additional complexity. The following section reviews psycholinguistic evidence supporting a view where anaphora retrieves representations of model objects rather than semantic forms, and explores implications for intensionally unspecific anaphora.

5 Processing anaphora

This section reviews psycholinguistic research on the cognitive mechanisms involved in understanding anaphora during *sentence processing*, the process of building a representation of the structure and meaning of an utterance. Section 5.1 outlines background assumptions about lexical activation and anaphoric reactivation—the idea that lexical representations maintain a dynamic activation level reflecting their likelihood of being needed at any given time (Anderson 1976, 2007; Anderson and Schooler 1991). When a pronoun is encountered, we assume that a representation of its antecedent is temporarily reactivated, producing observable effects of semantic priming. Section 5.2 discusses what exactly gets reactivated when a pronoun is processed, connecting to a contrast between descriptive and dynamic approaches. The descriptive approach assumes that anaphoric dependencies are established by retrieving a representation of the antecedent’s semantic form, dynamic approaches target its semantic value. Studies like Love and Swinney 1995 indicate that certain effects associated with lexical access are missing when interpreting anaphora, suggesting that the reactivated representation is meaning-based rather than form-based. Section 5.3 surveys mental- and situation-model theories (Johnson-Laird 1983; van Dijk and Kintsch 1983) which assume that discourse comprehension integrates utterance meaning with contextual inferences and world knowledge to construct a mental representation of described situations. This perspective supports a theory of meaning-based discourse representations relative to which anaphora are interpreted, connecting to dynamic semantic theories (see Boella and Lesmo 2002; Brasoveanu and Dotlačil 2020). Section 5.4 summarizes preliminary insights about how hypothetical contexts and intensional operators affect anaphoric processing.

5.1 Lexical activation and anaphoric reactivation

Many models of word recognition assume that lexical items are *activated* in working memory to be integrated into syntactic and semantic representations. The activation level of an item is an abstract quantity assumed to explain how accurately and quickly humans respond in various experimental tasks that require word recognition. By assumption, activation reflects the probability that the item will be needed at a given moment—an idea grounded in the notion that the organization of our mind and memory efficiently adapts to the environment and therefore mirrors its statistical properties (Anderson and Schooler 1991; Anderson 2007).

For example, higher-frequency words are assumed to have higher *base activation*, making them easier to activate enough to be manipulated in working memory. This assumption can explain the frequency effect—the robust and often replicated finding that high-frequency words elicit faster or more accurate responses than lower-frequency words, for instance in lexical decision tasks, where participants read a string of letters and indicate whether or not it is a word in their language (Staners et al. 1975; Whaley 1978). Similarly, in the visual world paradigm, where participants’ eye movements are recorded while listening to auditory stimuli, frequency effects have also been found, for instance, participants focus more on images of objects with more frequent names (Dahan et al. 2001).

Another common assumption is that memory is associative, allowing activation to spread from one item to semantically related ones. This explains the semantic priming effect, where accessing a word like *tiger* is facilitated by previous exposure to a semantically related word, like *lion*. This

effect is reflected, for instance, in lexical decision tasks (Meyer and Schvaneveldt 1971; Swinney et al. 1979) with faster response times when the target follows a semantically related prime. Priming effects typically last for 250–700 milliseconds, about three syllables (Nicol and Swinney 2003).

Because semantic priming provides a way to test for current activation in working memory, it has been used to investigate anaphora. A classic example is Nicol 1988, which suggests that processing anaphora involves reactivation of the antecedent. Participants in her study listened to sentences like (80), and perform a lexical decision task at the point marked by asterisk:

- (80) Nicol and Swinney 2003: (9)
*The ballerina told the skier that the doctor for the team would blame him *for the recent injury.*

When the target was semantically related to the antecedent (e.g., *snow* for *skier*), responses were faster than when it was related to another word in the sentence (e.g., *dance* for *ballerina*). This suggests that encountering the pronoun *him* reactivated the antecedent representation (see also, e.g., McElree and Bever 1989; MacDonald and MacWhinney 1990).

5.2 What is reactivated?

A central question is whether the reactivated representation corresponds to the full, form-based lexical entry or to an abstract, purely semantic representation.

Love and Swinney (1995) addressed this using cross-modal priming with a lexical decision task to test whether pronouns reactivated all lexical meanings of an ambiguous antecedent (e.g., *bank*) or only the contextually relevant one (see Nicol and Swinney 2003). Participants heard sentences like (81) and they saw a target word for lexical decision at one of three test points: immediately after the ambiguous word (*bank*), after a brief delay, or immediately following the pronoun (*it*).

- (81) *Jeff had read about problems with savings and loan institutions, so he went to the bank to ask about the safety that it provided with respect to CD investments.*

At the first test point they found priming effects for words related to both meanings (e.g., *financial institution* and *river edge*), consistent with well-established findings that all meanings of a lexically ambiguous word are initially activated when the word is heard in a sentence (e.g. Swinney 1979, Tanenhaus, Leiman, and Seidenberg 1979). At the second test point, no priming effects were found, consistent with the typical priming duration of 250–700 milliseconds. At the third and critical test point, immediately following the pronoun, a significant facilitation effect was found for only the target word related to the contextually relevant meaning, but not for the other meaning. This suggests that pronoun resolution selectively reactivated a semantic representation of the antecedent, rather than its full, form-based, lexical entry.

Finally, reading times for anaphoric pronouns show no facilitation for antecedents that are more frequent lexical items (Simner and Smyth 1999). In fact, van Gompel and Majid 2004 found the opposite effect—that pronouns with less frequent antecedents are read faster than those with more frequent ones. They suggest that infrequent nouns are allotted more attention during sentence comprehension and encoded into memory more strongly. Importantly, their results conflict with assuming full reaccess of a form-based lexical representation of the antecedent, which predict facilitation for more frequent lexical items.

The findings reviewed here provide converging evidence that pronoun resolution reactivates a more abstract semantic representation of the antecedent rather than a form-based representation.

5.3 Mental models

Theories of mental models (Johnson-Laird 1983, 2004; Johnson-Laird et al. 1989) and situation models (van Dijk and Kintsch 1983; Zwaan and Radvansky 1998) propose that discourse comprehension involves constructing a representation of described situations by integrating literal meaning with contextual inferences and world knowledge. These *mental models* are representations of objects and relations invoked in a discourse, continuously updated with new linguistic input and world knowledge (see Bara et al. 2001 for an implementation). Empirical support for this assumption comes from studies like MacDonald and Just 1989. In their memory-probe task, participants read sentences containing indefinite DPs with or without negation, like (82):

- (82) *Almost every weekend, Elizabeth bakes...*
- a. *no bread but only cookies for the children.*
 - b. *some bread but no cookies for the children.*
 - c. *some bread and some cookies for the children.*

When asked whether a visually presented target word (e.g. *bread* or *cookies*) had appeared in the previous sentence, participants responded faster to targets introduced without negation, independent of their position in the sentence, suggesting that non-negated referents had higher activation. This difference can be explained by assuming that these are integrated into a mental model of the described factual situation, increasing their activation. This aligns with converging evidence that semantic processing relies on an extensional referential representation, possibly in addition to form-based representations (e.g., Bransford et al. 1972; Johnson-Laird 1983; van Dijk and Kintsch 1983; Zwaan and Radvansky 1998; Garnham 2000; Kaup and Zwaan 2003).

Because processing anaphora pronouns appears to primarily retrieve semantic information, researchers have suggested that mental models provide an effective framework for anaphora processing (see, e.g., van Dijk and Kintsch 1983; Garnham 2000; Boella et al. 1999; Boella and Lesmo 2001, 2002; Brasoveanu and Dotlačil 2020). For example, van Dijk and Kintsch 1983 argued that extensional mental models explain referential continuity better than purely linguistic representations, a point Zwaan and Radvansky 1998 illustrate with (83):

- (83) Zwaan and Radvansky 1998, p. 164
- Lamar Alexander was behind in the polls. However, the former Tennessee governor remained optimistic. He considered it likely that a moderate candidate with new ideas would win the Republican nomination.*

The DPs *Lamar Alexander*, *the former governor of Tennessee*, *he* and *a moderate candidate with new ideas* all refer to the same individual. While the definite DP grammatically signals referential continuity, the indefinite would typically provide cues to introduce a new entity. Yet, comprehenders use domain knowledge and contextual inference to infer coreference, suggesting that mental models integrate world knowledge to track referents across discourse.

Mental models parallel dynamic semantic representations of discourse, where discourse referents and context updates capture how sentences introduce and modify referential possibilities.

Boella et al. (1999) and Boella and Lesmo (2001, 2002) illustrated that some key phenomena, like donkey- and plural anaphora, can also be analyzed using mental models. They suggest this offers cognitively plausible extensional representations of finite and limited size. Brasoveanu and Dotlačil (2020) further suggest that the mental-models approach is essentially equivalent to the model-theoretic interpretation of DRT and other dynamic systems (e.g., Sections 2.4 and 4), while dynamic semantics offers a rigorously developed mathematical foundation for representations and logical operations. There are some additional differences, for example, the model-theoretic interpretation in formal semantics does not assume the incorporation of relevant world knowledge. Further, it assumes a toy representation of concrete objects, while mental models assumes that referents and entities in the models are represented by abstract tokens, i.e., abstract of the instantiations of the mentioned entities and predicates that are quantified over in a discourse (see e.g., Johnson-Laird and Garnham 1980). However, an important parallel is that they are denotational representations of entities and referents that can be referenced in discourse interpretation. Arguments for mental-model based views of discourse interpretation thus push towards a value-based theory of anaphora, where anaphoric dependencies are established on a referential level (even an abstract one) between semantic values rather than descriptions, which are linguistic representations.

Finally, although mental-model theory assumes that the representations we construct when communicating about the world resemble those we form when perceiving it, it has been extended to hypothetical and counterfactual reasoning (e.g., Byrne 1997, 2005; Khemlani et al. 2018). Similarly, research on processing sentences with negation has argued that this involves mentally simulating the counterfactual scenario before shifting attention to the actual situation (e.g., Zwaan et al. 2002; Kaup and Zwaan 2003; Kaup et al. 2006, 2007). Tracking whether an entity belongs to a factual hypothetical, or counterfactual mental model thus connects naturally to the dynamic analyses of intensionally unspecific anaphora in Section 4. Meanwhile, the processing of hypothetical and counterfactual anaphora remains underexplored. The next section reviews some experimental evidence on this topic.

5.4 Intensionally unspecific antecedents

This section reviews experimental evidence on processing anaphora when the (potential) antecedents are intensionally unspecific. Section 5.4.1 addresses findings from factual anaphora where intensionally unspecific antecedents are unavailable, showing preliminary evidence that indefinites in hypothetical discourse contexts lead to difficulty that may be associated with a discourse-structural mismatch, and negated indefinites are not even considered as potential antecedents when they are inaccessible. Section 5.4.2 surveys limited previous work on processing acceptable uses of intensionally unspecific anaphora. Here, the findings are compatible with a difficulty associated with modal subordination in hypothetical anaphora. However, the processing of anaphora and their antecedents in hypothetical and counterfactual discourse is understudied.

5.4.1 Inaccessible antecedents

Dwivedi (1996) used sentence-by-sentence self-paced reading (SPR) to investigate how factual pronouns with hypothetical antecedents affect reading times. She compared four-sentence discourses like (84) where anaphoric pronouns were embedded under *would* (S3a) to unembedded factual pronouns (S3b).

- (84) Dwivedi 1996: (23)
 S1: *Cathy will probably buy a house.*
 S2: *She's been looking at the market lately.*
 S3a: *It would be a Colonial.* (hypothetical)
 S3b: *It's a Colonial.* (factual)
 S4: *Luckily, she has a good agent.*

Reading times were slower for sentences with factual pronouns, consistent with the observation that they are incompatible with hypothetical antecedents. In subsequent EEG-studies, Dwivedi et al. (2006, 2010) investigated the neurophysiological response to this unacceptability. In a 2×2 design, participants encountered a factual or hypothetical context (85), followed by a factual or hypothetical continuation (86).

- (85) Context sentence:
 a. *John is reading a novel.* (factual)
 b. *John is considering writing a novel.* (hypothetical)
- (86) Target sentence:
 a. *It ends quite abruptly.* (factual)
 b. *It might end quite abruptly.* (hypothetical)

The study analyzed event-related potentials (ERPs) at the pronoun (*it*), the verb (*ends/might*) and the following word. A key finding was a slow frontal positivity at the verb in the unacceptable hypothetical-factual condition, resembling a P600-like effect with a frontal distribution. Based on proposals linking frontal positivities to syntactic or structural revision, this may indicate a process of discourse-structural revision, in contrast to the N400 effect found for semantic conceptual anomalies like *John ate #democracy*.

Schmitz et al. (2024) investigated how negated indefinites affect memory retrieval during pronoun resolution (following Van Dyke 2007; Wagers et al. 2009; Kush et al. 2015). They built on studies like Badecker and Straub 2002, which examined sentences like (87), where a pronoun (*him*) appears with an accessible antecedent (*John*) and an inaccessible one (*Bill/Beth*) that is ruled out by Principle B of binding theory (e.g., to refer to *Bill*, the pronoun would have to be a reflexive).

- (87) Badecker and Straub 2002: (9–10)
 a. Multiple match
John thought that Bill owed him another chance to solve the problem.
 b. Accessible match
John thought that Beth owed him another chance to solve the problem.

Reading times following the pronoun were slower when multiple DPs matched its gender (87a) than when only the accessible antecedent did (87b). This suggests that pronoun resolution initially considers all gender-matching candidates, before selecting the accessible one. Interference arises when the cost of ruling out structurally inaccessible options causes a slowdown.

Schmitz et al. (2024) used a combination of offline measures and eye-tracking while reading to test whether matching gender features lead to interference effects for possible antecedents that are inaccessible due to negation in Dutch (like *geen zoon*, no son).

(88) [Schmitz et al. 2024](#): (17a,b)

a. Multiple match

De professor_m heeft geen zoon_m. De laatste jaren werkte hij_m...

The professor has no son. The last few years, he worked...

b. Accessible match

De professor_m heeft geen dochter_f. De laatste jaren werkte hij_m...

The professor has no daughter. The last few years, he worked...

...helaas op alle feestdagen.

...unfortunately during all holidays.

Unlike many previous studies, where structurally inaccessible items still interfere, Schmitz and colleagues found no interference from negated DPs, suggesting that they are not even considered as potential antecedents. They did observe weak evidence for an effect on right-bounded reading times in post-critical regions, which could suggest a limited role for inaccessible antecedents in resolving factual anaphora. The fact that negated antecedents can in principle support (hypothetical) anaphora suggests that these antecedents cannot be excluded from the search space for all anaphora. However, the literature on processing hypothetical anaphora is severely limited. Below, I summarize only a few studies that provide some very preliminary results.

5.4.2 Intensionally unspecific anaphora

While no study has directly investigated the processing of intensionally unspecific anaphora, this section reviews studies addressing some related questions.

Experiment 2 in [Dwivedi 1996](#) used sentence-by-sentence SPR to compare reading times for sentences with pronouns embedded under *will*, *should*, or *would* (89)–(90). Antecedents were proper names in sentences with and without *maybe*.

(89) [Dwivedi 1996](#): (27)

a. Susan will major in math. She will take Calculus III first. (F-F)

b. Susan will major in math. She should take Calculus III first. (F-NF)

c. Maybe Susan will major in math. She should take Calculus III first. (NF-NF)

(90) [Dwivedi 1996](#): (28)

a. Mario will run for President again. He will win this time. (F-F)

b. Mario will run for President again. He would win this time. (F-NF)

c. Maybe Mario will run for President again. He would win this time. (NF-NF)

They found a slowdown for target sentences with *would*, compared to those with *will* or *should*, which showed no slowdown. The *would*-slowdown occurred regardless of whether the antecedent sentence contained *maybe*. Because the antecedent DPs are proper names (i.e. factual definites), the effect might very well reflect a difficulty associated with the hypothetical modal *would*, independent of the anaphoric dependency.

Finally, related work examined anaphoric definites with hypothetical antecedents. [Haviland and Clark \(1974\)](#) showed that definites (*the alligator*) were read slower if the previous sentence introduced the antecedent in a hypothetical desire context (e.g., *Ed wanted an alligator for his birthday*), rather than explicitly asserting its existence (e.g., *Ed was given an alligator...*). They

argued that this cost suggests that comprehenders must accommodate an entity not established in the discourse model. Later replication attempts produced mixed results. Claus (2008) found a slowdown for German anaphoric definites with hypothetical antecedents in desire-contexts, while Delogu (2009) and Delogu et al. (2010) did not replicate the finding for Italian.

5.5 Interim conclusion

The research summarized in this section converges on the idea that pronoun resolution involves re-activation of extensional referents, rather than linguistic form. Cross-modal priming studies (Nicol 1988; Nicol and Swinney 2003, Section 5.1) show that hearing a pronoun briefly reactivates an antecedent representation, creating measurable facilitation for semantically related items. However, there is little evidence that this facilitation interacts with lexical properties like frequency effects and homophonous meanings (Section 5.2). Section 5.3 reviewed ideas from the mental models and situation-model frameworks, which further indicate that discourse comprehension relies on constructing an integrated extensional semantic representation of the situations described in a discourse. These representations track the status of each referent, capturing whether it is part of the real-world scenario, negated, or introduced under a modal operator, offering a connection to the dynamic analyses of intensionally unspecific anaphora in Section 4. However, to date, there is only limited evidence about the processing of anaphora in hypothetical and counterfactual discourse, summarized in Section 5.4.

6 Conclusion

This chapter has explored intensionally unspecific anaphora and their implications for understanding the semantics of anaphora from the perspectives of semantic theory, philosophical approaches to reference, and cognitive mechanisms involved in their interpretation.

The theoretical discussion traced how descriptive and dynamic approaches account for discourse anaphora. Descriptive approaches view anaphora as a dependency over semantic form, where pronouns recover the descriptive content of their antecedents, introducing presuppositions that must be satisfied or accommodated. This approach naturally extends to intensionally unspecific anaphora and it aligns with traditional externalist views of reference. In contrast, dynamic approaches capture anaphora as a dependency over semantic values, and uses drefs to provide a more comprehensive approach to anaphora across domains, including temporal and propositional anaphora. The challenge for dynamic semantics, however, has been to accommodate intensional and counterfactual anaphora, requiring extending the notion of drefs to track their dependencies on possible worlds.

The chapter has also reviewed experimental studies investigating cognitive mechanisms involved in interpreting anaphora. Psycholinguistic research suggests that this process involves semantic rather than lexical reactivation of antecedents, supporting the view that anaphoric dependencies are meaning-based rather than form-based. Research about lexical activation, mental models, and discourse representation indicate that anaphora is processed relative to situation models, where linguistic meaning is integrated with contextual knowledge. The findings tentatively suggest a dynamic semantic view of discourse processing, suggesting that language users track referents across possible contexts, including hypothetical and counterfactual scenarios.

Despite these insights, many questions remain. The processing of anaphora in intensional and counterfactual contexts has received limited empirical attention, and further experimental research is needed to explore how comprehenders establish and maintain reference in hypothetical and counterfactual discourse. The challenges posed by unspecific antecedents provide a testing ground for evaluating theoretical explanations, while empirical research continues to refine our understanding of the cognitive mechanisms underlying anaphoric reference. By integrating insights from formal semantics and psycholinguistics, the study of anaphora can further refine our understanding of anaphora and its role in human language comprehension.

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A Predicate logic

A.1 Semantics

- (1) Basic expressions
 - a. $\llbracket v \rrbracket^{M,g} = g(v)$ (for a variable v)
 - b. $\llbracket \alpha \rrbracket^{M,g} = I_M(\alpha)$ (for a non-logical constant α)
- (2) Atomic formulas
 - a. Predication:
 $\llbracket \pi(\alpha_1, \dots, \alpha_n) \rrbracket^{M,g} = 1$ iff $\langle \llbracket \alpha_1 \rrbracket^{M,g}, \dots, \llbracket \alpha_n \rrbracket^{M,g} \rangle \in \llbracket \pi \rrbracket^{M,g}$
 (for $n \in \mathbb{N}$, an n -ary predicate π , a sequence of n individual terms $\alpha_1, \dots, \alpha_n$)
- (3) Unary operators (for a formula ϕ)
 - a. Negation: $\llbracket \neg\phi \rrbracket^{M,g} = 1$, iff $\llbracket \phi \rrbracket^{M,g} = 0$
- (4) Binary operators (for formulas ϕ, ψ)
 - a. Conjunction
 $\llbracket \phi \wedge \psi \rrbracket^{M,g} = 1$, iff $\llbracket \phi \rrbracket^{M,g} = 1$ **and** $\llbracket \psi \rrbracket^{M,g} = 1$
 - b. Disjunction
 $\llbracket \phi \vee \psi \rrbracket^{M,g} = 1$, iff $\llbracket \phi \rrbracket^{M,g} = 1$ **or** $\llbracket \psi \rrbracket^{M,g} = 1$
 - c. Implication
 $\llbracket \phi \rightarrow \psi \rrbracket^{M,g} = 1$, iff $\llbracket \phi \rrbracket^{M,g} = 0$ **or** $\llbracket \psi \rrbracket^{M,g} = 1$
- (5) Lambda abstraction (for variable v , formula ϕ)
 $\llbracket \lambda v. \phi \rrbracket^{M,g} = \{ \langle d \in D_M, \llbracket \phi \rrbracket^{M,g[v \mapsto d]} \rangle \}$
- (6) Quantification (for variable v , formula ϕ)
 - a. Existential
 $\llbracket \exists v \phi \rrbracket^{M,g} = 1$, iff for some individual $d \in D_M$: $d \in \llbracket \lambda v. \phi \rrbracket^{M,g}$
 - b. Universal
 $\llbracket \forall v \phi \rrbracket^{M,g} = 1$, iff for all individuals $d \in D$: $d \in \llbracket \lambda v. \phi \rrbracket^{M,g}$
- (7) Definite operator (complex term, for a predicate P)
 $\llbracket \iota P \rrbracket^{M,g} = d$, if $d \in \llbracket P \rrbracket$, and $|\llbracket P \rrbracket| = 1$

A.2 Representations

- (8) *Every actress_x believes that she_x deserves more roles.* $\rightsquigarrow$
 $\forall x(\text{actress}(x) \rightarrow \text{believes}(x, \text{deserves-more-roles}(x)))$
- (9) *Mary owns a car_x.* $\rightsquigarrow \exists x(\text{car}(x) \wedge \text{own}(m, x))$
- (10) *It_x is red.* $\rightsquigarrow \text{red}(\iota P)$, where $P = \lambda x.(\text{car}(x) \wedge \text{own}(m, x))$
- (11) *Every farmer who owns [a donkey]_x feeds it_x.* $\rightsquigarrow$
 $\forall y((\text{farmer}(y) \wedge \exists x(\text{donkey}(x) \wedge \text{own}(y, x))) \rightarrow \text{feed}(y, \iota P))$,
 where $P = \lambda x.(\text{donkey}(x) \wedge \text{own}(y, x))$

B Dynamic predicate logic

B.1 Semantics

- (1) Basic expressions
 - a. $\llbracket v \rrbracket^{M,g} = g(v)$ (for a variable v)
 - b. $\llbracket \alpha \rrbracket^{M,g} = I_M(\alpha)$ (for a non-logical constant α)
- (2) Atomic formulas
 - a. Variable update
 $\llbracket [x] \rrbracket^{M,g} = \{h \mid \exists d \in D: h = g[x \mapsto d]\}$ (for an individual variable x)
 - b. Predication
 $\llbracket [\pi(\alpha_1, \dots, \alpha_n)] \rrbracket^{M,g} = \{h \mid h = g, \langle \llbracket \alpha_1 \rrbracket^{M,h}, \dots, \llbracket \alpha_n \rrbracket^{M,h} \rangle \in \llbracket \pi \rrbracket^{M,h}\}$
 (for $n \in \mathbb{N}$, an n -ary predicate π , a sequence of n individual terms $\alpha_1, \dots, \alpha_n$)
- (3) Unary operators (for a formula ϕ)
 - a. Negation
 $\llbracket [\neg\phi] \rrbracket^{M,g} = \{h \mid h = g, \llbracket \phi \rrbracket^{M,g} = \emptyset\}$ (for a formula ϕ)
- (4) Binary connectives (for formulas ϕ, ψ)
 - a. Dynamic conjunction, i.e. relation composition
 $\llbracket [\phi; \psi] \rrbracket^{M,g} = \{h \mid \exists i \in \llbracket \phi \rrbracket^{M,g}: h \in \llbracket \psi \rrbracket^{M,i}\}$
 - b. Disjunction
 $\llbracket [\phi \vee \psi] \rrbracket^{M,g} = \{h \mid h = g, \llbracket \phi \rrbracket^{M,g} \cup \llbracket \psi \rrbracket^{M,g} \neq \emptyset\}$
 - c. Implication
 $\llbracket [\phi \rightarrow \psi] \rrbracket^{M,g} = \{h \mid h = g, \forall i \in \llbracket \phi \rrbracket^{M,g}: \llbracket \psi \rrbracket^{M,i} \neq \emptyset\}$
- (5) Universal quantification
 $\llbracket [\forall v. \phi] \rrbracket^{M,g} = \{h \mid h = g, \forall d \in D: \llbracket \phi \rrbracket^{M,g[v \mapsto d]} \neq \emptyset\}$
- (6)
 - a. Update
 $C + S := \bigcup_{g \in C} \llbracket S \rrbracket^{M,g}$
 - b. Truth
 S is true relative to model M and context C , iff $\bigcup_{g \in C} \llbracket S \rrbracket^{M,g} \neq \emptyset$

B.2 Derivations for updates

- (7) It_x is red.: $\llbracket \text{red}(x) \rrbracket^{M,g} = \{h \mid h = g, h(x) \in I_M(\text{red})\}$ (2b), (1)
- (8) *Mary owns a car_x*.
 - a. $\llbracket [x]; [\text{car}(x)]; [\text{own}(m, x)] \rrbracket^{M,g} =$ (4a), (2a)
 - b. $\{h \mid \exists d \in D: h \in \llbracket [\text{car}(x)]; [\text{own}(m, x)] \rrbracket^{M,g[x \mapsto d]}\} =$ (4a), (1)
 - c. $\{g[x \mapsto d] \mid d \in I_M(\text{car}), \langle I_M(m), d \rangle \in I_M(\text{own})\}$
- (9) *Every_x actress believes that she_x deserves more roles*.
 - a. $\llbracket [\forall x([\text{actress}(x)] \rightarrow [\text{believes}(x, \text{deserve-more-roles}(x))])] \rrbracket^{M,g} =$ (5)

- b. $\{h \mid h = g, \forall d \in D: \llbracket \text{actress}(x) \rightarrow \text{believes}(x, \text{deserve-more-roles}(x)) \rrbracket^{M, g[x \mapsto d]} \neq \emptyset\} =$ (4c)
- c. $\{h \mid h = g, \forall d \in D: \forall k \in \llbracket \text{actress}(x) \rrbracket^{M, g[x \mapsto d]}: \llbracket \text{believes}(x, \text{deserve-more-roles}(x)) \rrbracket^{M, k} \neq \emptyset\}$
- (10) *Every_x farmer who owns a_y donkey feeds it_y.*
- a. $\llbracket \forall x ((\llbracket \text{farmer}(x) \rrbracket; \llbracket y \rrbracket; \llbracket \text{donkey}(y) \rrbracket; \llbracket \text{own}(x, y) \rrbracket) \rightarrow \llbracket \text{feed}(x, y) \rrbracket) \rrbracket^{M, g} =$ (5)
- b. $\{h \mid h = g, \forall d_1 \in D: \llbracket (\llbracket \text{farmer}(x) \rrbracket; \llbracket y \rrbracket; \llbracket \text{donkey}(y) \rrbracket; \llbracket \text{own}(x, y) \rrbracket) \rightarrow \llbracket \text{feed}(x, y) \rrbracket \rrbracket^{M, g[x \mapsto d_1]} \neq \emptyset\} =$ (4c)
- c. $\{h \mid h = g, \forall d_1 \in D: \forall i \in \llbracket (\llbracket \text{farmer}(x) \rrbracket; \llbracket y \rrbracket; \llbracket \text{donkey}(y) \rrbracket; \llbracket \text{own}(x, y) \rrbracket) \rrbracket^{M, g[x \mapsto d_1]}: \llbracket \llbracket \text{feed}(x, y) \rrbracket \rrbracket^{M, i} \neq \emptyset\} =$ (4a)
- d. $\{h \mid h = g, \forall d_1 \in D: \forall i \in \{j \mid \exists k \in \llbracket \llbracket \text{farmer}(x) \rrbracket \rrbracket^{M, g[x \mapsto d_1]}: j \in \llbracket (\llbracket y \rrbracket; \llbracket \text{donkey}(y) \rrbracket; \llbracket \text{own}(x, y) \rrbracket) \rrbracket^{M, k}\}: \llbracket \llbracket \text{feed}(x, y) \rrbracket \rrbracket^{M, i} \neq \emptyset\} =$ (2b), (1)
- e. $\{h \mid h = g, \forall d_1 \in D: \forall i \in \{j \mid d_1 \in I_M(\text{farmer}), j \in \llbracket (\llbracket y \rrbracket; \llbracket \text{donkey}(y) \rrbracket; \llbracket \text{own}(x, y) \rrbracket) \rrbracket^{M, g[x \mapsto d_1]}\}: \llbracket \llbracket \text{feed}(x, y) \rrbracket \rrbracket^{M, i} \neq \emptyset\} =$ (4a), (2a), (2b), (1)
- f. $\{h \mid h = g, \forall d_1 \in D: \forall i \in \{g[x \mapsto d_1, y \mapsto d_2] \mid d_1 \in I_M(\text{farmer}), d_2 \in I_M(\text{donkey}), \langle d_1, d_2 \rangle \in I_M(\text{own})\}: \llbracket \llbracket \text{feed}(x, y) \rrbracket \rrbracket^{M, i} \neq \emptyset\}$
- (11) *Mary doesn't own a_x car.* (3a)
- $\llbracket \llbracket \neg(\llbracket x \rrbracket; \llbracket \text{car}(x) \rrbracket; \llbracket \text{own}(m, x) \rrbracket) \rrbracket \rrbracket^{M, g} = \{h \mid h = g, \llbracket (\llbracket x \rrbracket; \llbracket \text{car}(x) \rrbracket; \llbracket \text{own}(m, x) \rrbracket) \rrbracket^{M, g} = \emptyset\}$
- (12) *It's not the case that Mary doesn't own a_x car.*
- a. $\llbracket \llbracket \neg \neg(\llbracket x \rrbracket; \llbracket \text{car}(x) \rrbracket; \llbracket \text{own}(m, x) \rrbracket) \rrbracket \rrbracket^{M, g} =$ (3a)
- b. $\{h \mid h = g, \llbracket \llbracket \neg \neg(\llbracket x \rrbracket; \llbracket \text{car}(x) \rrbracket; \llbracket \text{own}(m, x) \rrbracket) \rrbracket \rrbracket^{M, g} = \emptyset\} =$ (3a)
- c. $\{h \mid h = g, \{i \mid i = g, \llbracket (\llbracket x \rrbracket; \llbracket \text{car}(x) \rrbracket; \llbracket \text{own}(m, x) \rrbracket) \rrbracket^{M, g} = \emptyset\} = \emptyset\} =$
- d. $\{h \mid h = g, \llbracket (\llbracket x \rrbracket; \llbracket \text{car}(x) \rrbracket; \llbracket \text{own}(m, x) \rrbracket) \rrbracket^{M, g} \neq \emptyset\}$

C File change semantics with partially familiar drefs

C.1 Semantics

- (1) Basic expressions
- a. $\llbracket v \rrbracket^{M, w, g} = g(v)$ (for a variable v)
- b. $\llbracket \alpha \rrbracket^{M, w, g} = I_M(\alpha)(w)$ (for a non-logical constant α)
- (2) Atomic formulas
- a. Variable update
 $\llbracket [x] \rrbracket^{M, C} = \{\langle w, h \rangle \mid \langle w, g \rangle \in C, \exists d \in D: h = g[x \mapsto d]\}$
 (for an individual variable x)
- b. Predication
 $\llbracket \llbracket \pi(\alpha_1, \dots, \alpha_n) \rrbracket \rrbracket^{M, C} =$

- $\{\langle w, g \rangle \in C \mid \langle \llbracket \alpha_1 \rrbracket^{M,w,g}, \dots, \llbracket \alpha_n \rrbracket^{M,w,g} \rangle \in \llbracket \pi \rrbracket^{M,w,g}\}$
(for $n \in \mathbb{N}$, an intensional n -ary predicate π , a sequence of n individual terms $\alpha_1, \dots, \alpha_n$)
- c. Identity: $\llbracket [\alpha_1 = \alpha_2] \rrbracket^{M,C} = \{\langle w, g \rangle \in C \mid \llbracket \alpha_1 \rrbracket^{M,C} = \llbracket \alpha_2 \rrbracket^{M,C}\}$
(for terms α_1, α_2)
- (3) Unary operators (for a formula ϕ)
- a. Negation:
 $\llbracket [\neg \phi] \rrbracket^{M,C} = \{\langle w, g \rangle \in C \mid \llbracket \phi \rrbracket^{M,\{\langle w,g \rangle\}} = \emptyset\}$
- b. Epistemic possibility:
 $\llbracket [\Diamond \phi] \rrbracket^{M,C} = \{\langle w, g \rangle \in C \mid \llbracket \phi \rrbracket^{M,C} \neq \emptyset\}$
- c. Epistemic necessity (for an epistemic accessibility relation R):
 $\llbracket [\Box \phi] \rrbracket^{M,C} = \{\langle w, g \rangle \in C \mid \forall w' \in R(w) : \llbracket \phi \rrbracket^{M,\{\langle w',g \rangle\}} \neq \emptyset\}$
- d. Remote epistemic necessity
(for epistemic accessibility relation R , remote possibility B)
 $\llbracket [\text{would}(\phi)] \rrbracket^{M,C} = \{\langle w, g \rangle \in C \mid \forall w' \in (R(w) \cap B) : \llbracket \phi \rrbracket^{M,\{\langle w',g \rangle\}} \neq \emptyset\}$
- (4) Binary operators (for formulas ϕ, ψ)
- a. Dynamic conjunction, i.e. relation composition
 $\llbracket [\phi; \psi] \rrbracket^{M,C} = \llbracket \psi \rrbracket^{M, \llbracket \phi \rrbracket^{M,C}}$
- b. Disjunction
 $\llbracket [\phi \vee \psi] \rrbracket^{M,C} = \{\langle w, g \rangle \in C \mid \llbracket \phi \rrbracket^{M,\{\langle w,g \rangle\}} \cup \llbracket \psi \rrbracket^{M,\{\langle w,g \rangle\}} \neq \emptyset\}$
- c. Implication
 $\llbracket [\phi \rightarrow \psi] \rrbracket^{M,C} = \{\langle w, g \rangle \in C \mid \llbracket \phi \rrbracket^{M,\{\langle w,g \rangle\}} \subseteq \llbracket \psi \rrbracket^{M,\{\langle w,g \rangle\}}\}$
- d. Biconditional
 $\llbracket [\phi \leftrightarrow \psi] \rrbracket^{M,C} = \{\langle w, g \rangle \in C \mid \llbracket \phi \rrbracket^{M,\{\langle w,g \rangle\}} = \llbracket \psi \rrbracket^{M,\{\langle w,g \rangle\}}\}$
- (5) a. Update
 $C + S := \llbracket S \rrbracket^{M,C}$
- b. Truth
 S is true relative to model M and context C , iff $\llbracket S \rrbracket^{M,C} \neq \emptyset$
- (6) Existential quantification: $C[\exists x(\phi)]$
- a. Asserts: $C([x]; [\phi])$
- b. Projects: $\lambda C. C([x]; [\phi]) \cup C([x]; [\phi])$

C.2 Derivations

- (7) (Assertion of) Mary owns [a car]_x.
- a. $\llbracket [x]; [\text{car}(x)]; [\text{own}(m, x)] \rrbracket^{M,C} =$ (4a)
- b. $\llbracket [\text{own}(m, x)] \rrbracket^{M, \llbracket [\text{car}(x)] \rrbracket^{M, \llbracket [x] \rrbracket^{M,C}}} =$ (2b), (1)
- c. $\{\langle w, i \rangle \in \{\langle w', h \rangle \in \llbracket [x] \rrbracket^{M,C} \mid h(x) \in I_M(\text{car})(w')\} \mid \langle I_M(m), i(x) \rangle \in I_M(\text{own})(w)\} =$

- d. $\{\langle w, i \rangle \in \llbracket [x] \rrbracket^{M,C} \mid i(x) \in I_M(\text{car})(w), \langle I_M(m), i(x) \rangle \in I_M(\text{own})(w) \} =$ (2a)
- e. $\{\langle w, i \rangle \in \{\langle w', h \rangle \mid \langle w', g \rangle \in C, \exists d \in D: h = g[x \mapsto d]\} \mid$
 $i(x) \in I_M(\text{car})(w), \langle I_M(m), i(x) \rangle \in I_M(\text{own})(w) \} =$
- f. $\{\langle w, g[x \mapsto d] \rangle \mid \langle w, g \rangle \in C, d \in I_M(\text{car})(w), \langle I_M(m), d \rangle \in I_M(\text{own})(w) \}$
- (8) (Assertion of) Mary doesn't own [a car]_x
- a. $\llbracket \neg([x]; [\text{car}(x)]; [\text{own}(m, x)]) \rrbracket^{M,C} =$ (3a)
- b. $\{\langle w, g \rangle \in C \mid \llbracket ([x]; [\text{car}(x)]; [\text{own}(m, x)]) \rrbracket^{M, \{\langle w, g \rangle\}} = \emptyset \}$
- (9) (Assertion of) It's not the case that Mary doesn't own [a car]_x
- a. $\llbracket \neg \neg([x]; [\text{car}(x)]; [\text{own}(m, x)]) \rrbracket^{M,C} =$ (3a)
- b. $\{\langle w, g \rangle \in C \mid \llbracket \neg([x]; [\text{car}(x)]; [\text{own}(m, x)]) \rrbracket^{M, \{\langle w, g \rangle\}} = \emptyset \} =$ (3a)
- c. $\{\langle w, g \rangle \in C \mid \{\langle w', g' \rangle \in \{\langle w, g \rangle\} \mid$
 $\llbracket ([x]; [\text{car}(x)]; [\text{own}(m, x)]) \rrbracket^{M, \{\langle w', g' \rangle\}} = \emptyset \} = \emptyset \} =$
- d. $\{\langle w, g \rangle \in C \mid \llbracket ([x]; [\text{car}(x)]; [\text{own}(m, x)]) \rrbracket^{M, \{\langle w, g \rangle\}} \neq \emptyset \}$
- (10) It_x's red.
- a. $\llbracket [\text{red}(x)] \rrbracket^{M,C} =$ (2b), (1)
- b. $\{\langle w, g \rangle \in C \mid g(x) \in I_M(\text{red})(w) \}$
- (11) (Assertion of) Mary might own [a car]_x.
- a. $\llbracket [\Diamond([x]; [\text{car}(x)]; [\text{own}(m, x)])] \rrbracket^{M,C} =$ (3b)
- b. $\{\langle w, g \rangle \in C \mid \llbracket [x]; [\text{car}(x)]; [\text{own}(m, x)] \rrbracket^{M,C} \neq \emptyset \} =$ (7f)
- c. $\{\langle w, g \rangle \in C \mid \{\langle w', g'[x \mapsto d] \rangle \mid$
 $\langle w', g' \rangle \in C, d \in I_M(\text{car})(w'), \langle I_M(m), d \rangle \in I_M(\text{own})(w') \} \neq \emptyset \}$
- (12) It would be red.
- a. $\llbracket [\text{would}([\text{red}(x)])] \rrbracket^{M,C} =$ (3c)
- b. $\{\langle w, g \rangle \in C \mid \forall w' \in (R(w) \cap B) : \llbracket [\text{red}(x)] \rrbracket^{M, \{\langle w', g \rangle\}} \neq \emptyset \}$
- (13) (Assertion of) Either Mary doesn't own [a car]_x, or it_x's red.
- a. $\llbracket [(\neg([x]; [\text{car}(x)]; [\text{own}(m, x)]) \vee [\text{red}(x)])] \rrbracket^{M,C} =$ (4b)
- b. $\{\langle w, g \rangle \in C \mid \llbracket \neg([x]; [\text{car}(x)]; [\text{own}(m, x)]) \rrbracket^{M, \{\langle w, g \rangle\}} \cup \llbracket [\text{red}(x)] \rrbracket^{M, \{\langle w, g \rangle\}} \neq \emptyset \} =$
(8), (10b)
- c. $\{\langle w, g \rangle \in C \mid \{\langle w', g' \rangle \in \{\langle w, g \rangle\} \mid$
 $\{\langle w, g'[x \mapsto d] \rangle \mid d \in I_M(\text{car})(w'), \langle I_M(m), d \rangle \in I_M(\text{own})(w') \} = \emptyset \} \cup$
 $\{\langle w', g' \rangle \in \{\langle w, g \rangle\} \mid g'(x) \in I_M(\text{red})(w') \} \neq \emptyset \} =$
- d. $\{\langle w, g \rangle \in C \mid$
 $\{\langle w, g[x \mapsto d] \rangle \mid d \in I_M(\text{car})(w'), \langle I_M(m), d \rangle \in I_M(\text{own})(w) \} = \emptyset \} \cup$
 $\{\langle w, g \rangle \mid g(x) \in I_M(\text{red})(w) \} \neq \emptyset \}$

D Intensional logic of change

D.1 Semantics

- (1) Basic expressions
 - a. $\llbracket v \rrbracket^{M,w,g} = g(v)(w)$ (for a variable v)
 - b. $\llbracket \alpha \rrbracket^{M,w,g} = I_M(\alpha)(w)$ (for a non-logical constant α)
- (2) Atomic formulas
 - a. Relative variable update (individual concepts)

$$\llbracket [x] \rrbracket^{M,W,g} = \{h \mid \exists f: D_w \mapsto D_e, h = g[x \mapsto f], \\ \forall w \in D_w: w \in W \leftrightarrow x(w) \neq \#\}$$
 (for an individual concept variable x , of type $\langle w, e \rangle$)
 - b. Predication

$$\llbracket [\pi(\alpha_1, \dots, \alpha_n)] \rrbracket^{M,W,g} = \\ \{h \mid h = g, \forall w \in W: \langle \llbracket \alpha_1 \rrbracket^{M,w,h} \rangle \dots \langle \llbracket \alpha_n \rrbracket^{M,w,h} \rangle \in \llbracket \pi \rrbracket^{M,w,h}\}$$
 (for $n \in \mathbb{N}$, an intensional n -ary predicate π , a sequence of n individual terms $\alpha_1, \dots, \alpha_n$)
- (3) Unary operators (for a formula ϕ)
 - a. Negation

$$\llbracket [\neg \phi] \rrbracket^{M,W,g} = \{h \mid \exists W' \subseteq D_w: W' = \overline{W}, h \in \llbracket \phi \rrbracket^{M,W',g}\}$$
 - b. Epistemic possibility:

$$\llbracket [\Diamond \phi] \rrbracket^{M,W,g} = \{h \mid \exists W' \subseteq D_w: W \cap W' \neq \emptyset, h \in \llbracket \phi \rrbracket^{M,W',g}\}$$
 - c. Remote necessity:

$$\llbracket [\text{would } \phi] \rrbracket^{M,W,g} = \{h \mid \exists W' \subseteq D_w: h \in \llbracket \phi \rrbracket^{M,W',g}\}$$
 - d. $\llbracket [\text{would } ([\text{red}(x)])] \rrbracket^{M,W_1,g} = \\ \{h \mid \exists W_2 \subseteq D_w: h \in \llbracket [\text{red}(x)] \rrbracket^{M,W_2,g}, \forall w \in W_1: (R(w) \cap B) \subseteq W_2\}$
- (4) Binary operators (for formulas ϕ, ψ)
 - a. Dynamic conjunction, i.e. relation composition

$$\llbracket [\phi; \psi] \rrbracket^{M,W,g} = \{h \mid \exists i \in \llbracket \phi \rrbracket^{M,W,g}: h \in \llbracket \psi \rrbracket^{M,W,i}\}$$
 - b. Disjunction

$$\llbracket [\phi \vee \psi] \rrbracket^{M,W,g} = \\ \{h \mid \exists W' \subseteq D_w, W'' \subseteq D_w: W = W' \cup W'', \exists i \in \llbracket \phi \rrbracket^{M,W',g}: h \in \llbracket \psi \rrbracket^{M,W'',i}\}$$
 - c. Implication:

$$\llbracket [\phi \rightarrow \psi] \rrbracket^{M,W,g} = \{h \mid \exists W' \subseteq D_w, W'' \subseteq D_w: W = W' \cup W'', W' \subseteq W'', \\ \exists i \in \llbracket \phi \rrbracket^{M,W',g}: h \in \llbracket \psi \rrbracket^{M,W'',i}\}$$
- (5)
 - a. Context: $C = \{\langle W, g \rangle\}$
 - b. Update: $C + S := \bigcup_{\langle W, g \rangle \in C} \{\langle W \cap W', h \rangle \mid h \in \llbracket S \rrbracket^{M,W',g}\}$
 - c. Truth
 S is true relative to model M and context C , iff $\exists \langle W, g \rangle \in C + S: W \neq \emptyset$

D.2 Derivations

(6) Mary owns [a car]_x

$$a. \llbracket ([x]; [\text{car}(x)]; [\text{own}(\mathbf{m})(x)] \rrbracket^{M,W,g} = \quad (4a)$$

$$b. \{h \mid \exists i \in \llbracket [x] \rrbracket^{M,W,g} : \exists j \in \llbracket [\text{car}(x)] \rrbracket^{M,W,i} : h \in \llbracket [\text{own}(\mathbf{m})(x)] \rrbracket^{M,W,j} \} = \quad (2a)$$

$$c. \{h \mid \exists i \in \{k \mid \exists f : D_w \mapsto D_e, k = g[x \mapsto f], \forall w \in D_w : w \in W \leftrightarrow x(w) \neq \#\} : \exists j \in \llbracket [\text{car}(x)] \rrbracket^{M,W,i} : h \in \llbracket [\text{own}(\mathbf{m})(x)] \rrbracket^{M,W,j} \} =$$

$$d. \{h \mid \exists f : D_w \mapsto D_e, \forall w \in D_w : w \in W \leftrightarrow x(w) \neq \#, \exists j \in \llbracket [\text{car}(x)] \rrbracket^{M,W,g[x \mapsto f]} : h \in \llbracket [\text{own}(\mathbf{m})(x)] \rrbracket^{M,W,j} \} = \quad (2b), (1)$$

$$e. \{h \mid \exists f : D_w \mapsto D_e, \forall w \in D_w : w \in W \leftrightarrow x(w) \neq \#, \forall w \in W : f(w) \in I_M(\text{car})(w), h \in \llbracket [\text{own}(\mathbf{m})(x)] \rrbracket^{M,W,g[x \mapsto f]} \} = \quad (2b), (1)$$

$$f. \{g[x \mapsto f] \mid f : D_w \mapsto D_e, \forall w \in D_w : w \in W \leftrightarrow x(w) \neq \#, \forall w \in W : f(w) \in I_M(\text{car})(w), \forall w \in W : \langle I_M(\mathbf{m}), k(x)(w) \rangle \in I_M(\text{own})(w) \}$$

(7) Mary doesn't own a car

$$a. \llbracket \neg([x]; [\text{car}(x)]; [\text{own}(\mathbf{m})(x)] \rrbracket^{M,W,g} = \quad (3a)$$

$$b. \{h \mid \exists W' \subseteq D_w : W' = \overline{W}, h \in \llbracket [x]; [\text{car}(x)]; [\text{own}(\mathbf{m})(x)] \rrbracket^{M,W',g} \} = \quad (6f)$$

$$c. \{h \mid \exists W' \subseteq D_w : W' = \overline{W}, h \in \{g[x \mapsto f] \mid f : D_w \mapsto D_e, \forall w \in D_w : w \in W' \leftrightarrow x(w) \neq \#, \forall w \in W' : f(w) \in I_M(\text{car}), \forall w \in W' : \langle I_M(\mathbf{m}), k(x)(w) \rangle \in I_M(\text{own})(w) \}$$

$$d. \{g[x \mapsto f] \mid \exists W' \subseteq D_w : W' = \overline{W}, f : D_w \mapsto D_e, \forall w \in D_w : w \in W' \leftrightarrow x(w) \neq \#, \forall w \in W' : f(w) \in I_M(\text{car}), \forall w \in W' : \langle I_M(\mathbf{m}), k(x)(w) \rangle \in I_M(\text{own})(w) \}$$

(8) It_x's red.

$$a. \llbracket [\text{red}(x)] \rrbracket^{M,W,g} = \quad (2b), (1)$$

$$b. \{h \mid h = g, \forall w \in W : h(x)(w) \in I_M(\text{red})(w) \}$$

(9) It_x would be red.

$$a. \llbracket [\text{would}([\text{red}(x)])] \rrbracket^{M,W,g} =$$

$$b. \{h \mid \exists W' \subseteq D_w : h \in \llbracket [\text{red}(x)] \rrbracket^{M,W',g}, \forall w \in W : (R(w) \cap B) \subseteq W' \}$$

(10) It's not the case that Mary doesn't own a car

$$a. \llbracket \neg \neg([x]; [\text{car}(x)]; [\text{own}(\mathbf{m})(x)] \rrbracket^{M,W,g} = \quad (3a)$$

$$b. \{h \mid \exists W' \subseteq D_w : W' = \overline{W}, h \in \llbracket \neg([x]; [\text{car}(x)]; [\text{own}(\mathbf{m})(x)] \rrbracket^{M,W',g} \} = \quad (7d)$$

$$c. \{h \mid \exists W' \subseteq D_w : W' = \overline{W}, h \in \{g[x \mapsto f] \mid \exists W'' \subseteq D_w : W'' = \overline{W'}, f : D_w \mapsto D_e, \forall w \in D_w : w' \in W'' \leftrightarrow x(w) \neq \#, \forall w \in W'' : f(w) \in I_M(\text{car}), \forall w \in W'' : \langle I_M(\mathbf{m}), k(x)(w) \rangle \in I_M(\text{own})(w) \} \} =$$

- d. $\{g[x \mapsto f] \mid \exists W' \subseteq D_w : W' = \overline{W}, \exists W'' \subseteq D_w : W'' = \overline{W'},$
 $f: D_w \mapsto D_e, \forall w \in D_w: w' \in W' \leftrightarrow x(w) \neq \#,$
 $\forall w \in W'': f(w) \in I_M(\text{car}),$
 $\forall w \in W'': \langle I_M(m), k(x)(w) \rangle \in I_M(\text{own})(w)\}$

(11) Either Mary doesn't own [a car]_x, or it_x's red.

a. $\llbracket [(\neg([x]; [\text{car}(x)]; [\text{own}(m)(x)]) \vee [\text{red}(x)])] \rrbracket^{M, W, g} =$ (4b)

b. $\{h \mid \exists W' \subseteq D_w, W'' \subseteq D_w: W = W' \cup W'',$
 $\exists i \in \llbracket [(\neg([x]; [\text{car}(x)]; [\text{own}(m)(x)])] \rrbracket^{M, W', g}: h \in \llbracket [\text{red}(x)] \rrbracket^{M, W'', i}\}$ (8), (10b)

c. $\{h \mid \exists W' \subseteq D_w, W'' \subseteq D_w: W = W' \cup W'',$
 $\exists i \in \{g[x \mapsto f] \mid \exists W''' \subseteq D_w: W''' = \overline{W'},$
 $f: D_w \mapsto D_e, \forall w \in D_w: w \in W''' \leftrightarrow x(w) \neq \#,$
 $\forall w \in W''': f(w) \in I_M(\text{car}),$
 $\forall w \in W''': \langle I_M(m), k(x)(w) \rangle \in I_M(\text{own})(w)\}: h \in \{j \mid j = i, \forall w \in W'': j(x)(w) \in I_M(\text{red})(w)\}\} =$

d. $\{g[x \mapsto f] \mid \exists W' \subseteq D_w, W'' \subseteq D_w: W = W' \cup W'',$
 $\exists W''' \subseteq D_w: W''' = \overline{W'},$
 $f: D_w \mapsto D_e, \forall w \in D_w: w \in W''' \leftrightarrow x(w) \neq \#,$
 $\forall w \in W''': f(w) \in I_M(\text{car}),$
 $\forall w \in W''': \langle I_M(m), k(x)(w) \rangle \in I_M(\text{own})(w),$
 $\forall w \in W'': j(x)(w) \in I_M(\text{red})(w)\}$

(12) Mary might own [a car]_x.

a. $\llbracket [\Diamond([x]; [\text{car}(x)]; [\text{own}(m)(x)])] \rrbracket^{M, W, g} =$ (3b)

b. $\{h \mid \exists W' \subseteq D_w: W \cap W' \neq \emptyset, h \in \llbracket [x]; [\text{car}(x)]; [\text{own}(m)(x)] \rrbracket^{M, W', g}\}$ (6f)

c. $\{h \mid \exists W' \subseteq D_w: W \cap W' \neq \emptyset, h \in$
 $\{g[x \mapsto f] \mid f: D_w \mapsto D_e, \forall w \in D_w: w \in W' \leftrightarrow x(w) \neq \#,$
 $\forall w \in W': f(w) \in I_M(\text{car})(w),$
 $\forall w \in W': \langle I_M(m), k(x)(w) \rangle \in I_M(\text{own})(w)\}\} =$

d. $\{g[x \mapsto f] \mid \exists W' \subseteq D_w: W \cap W' \neq \emptyset,$
 $f: D_w \mapsto D_e, \forall w \in D_w: w \in W' \leftrightarrow x(w) \neq \#,$
 $\forall w \in W': f(w) \in I_M(\text{car})(w),$
 $\forall w \in W': \langle I_M(m), k(x)(w) \rangle \in I_M(\text{own})(w)\}$

D.3 Updates

D.3.1 Double negation

(13) a. It's not the case that Mary doesn't own a car

b. $\llbracket \neg\neg([x]; [\text{car}(x)]; [\text{own}(m)(x)]) \rrbracket^{M_\#, W, g} = \{g[x \mapsto f] \mid \exists W' \subseteq D_w: W' = \overline{W}, \exists W'' \subseteq$
 $D_w: W'' = \overline{W'},$
 $f: D_w \mapsto D_e, \forall w \in D_w: w' \in W' \leftrightarrow x(w) \neq \#,$
 $\forall w \in W'': f(w) \in I_M(\text{car}),$
 $\forall w \in W'': \langle I_M(m), k(x)(w) \rangle \in I_M(\text{own})(w)\}$

$$C_0 \longrightarrow \left\langle \{w_1, w_2\}, g[x \mapsto \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_2 \\ w_3 \mapsto \# \\ w_4 \mapsto \# \end{bmatrix}] \right\rangle$$

Figure 22: Update with *It's not the case that Mary doesn't own a car*

D.3.2 Bathroom sentences

- (14) a. Either Mary doesn't own [a car]_x, or it_x's red.
b. $\llbracket [(\neg([x]; [\text{car}(x)]); [\text{own}(\mathbf{m})(x)]) \vee [\text{red}(x)]] \rrbracket^{M, W, g} = \{g[x \mapsto f] \mid \exists W' \subseteq D_w, W'' \subseteq D_w : W = W' \cup W'', \\ \exists W''' \subseteq D_w : W''' = \overline{W'}, \\ f : D_w \mapsto D_e, \forall w \in D_w : w \in W''' \leftrightarrow x(w) \neq \#, \\ \forall w \in W''' : f(w) \in I_M(\text{car}), \\ \forall w \in W''' : \langle I_M(\mathbf{m}), k(x)(w) \rangle \in I_M(\text{own})(w), \\ \forall w \in W'' : j(x)(w) \in I_M(\text{red})(w)\}$

$$C_0 \longrightarrow \left\langle \{w_1, w_3, w_4\}, g[x \mapsto \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_2 \\ w_3 \mapsto \# \\ w_4 \mapsto \# \end{bmatrix}] \right\rangle$$

Figure 23: Update with *Either Mary doesn't own a car, or it's red.*

D.3.3 Modal subordination

- (15) a. Mary might own [a car]_x. It_x would be red
 b. $\llbracket [\Diamond([x]; [\text{car}(x)]; [\text{own}(\mathbf{m})(x)])] \rrbracket^{M, W, g} =$
 $\{g[x \mapsto f] \mid \exists W' \subseteq D_w : W \cap W' \neq \emptyset,$
 $f : D_w \mapsto D_e, \forall w \in D_w : w \in W' \leftrightarrow x(w) \neq \#,$
 $\forall w \in W' : f(w) \in I_M(\text{car})(w),$
 $\forall w \in W' : \langle I_M(\mathbf{m}), k(x)(w) \rangle \in I_M(\text{own})(w) \}$

$$C_0 \longrightarrow \left\langle \{w_1, w_2, w_3, w_4\}, g[x \mapsto \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_2 \\ w_3 \mapsto \# \\ w_4 \mapsto \# \end{bmatrix}] \right\rangle - \left\langle \{w_1, w_3\}, g[x \mapsto \begin{bmatrix} w_1 \mapsto c_1 \\ w_2 \mapsto c_2 \\ w_3 \mapsto \# \\ w_4 \mapsto \# \end{bmatrix}] \right\rangle$$

Figure 24: Update with *Mary might own a car. It would be red.*